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QUESTION BANK OF INTERNET OF EVERYTHING

B.E SEM-VII **SCHEME: R-19 C SCHEME** **FOR WINTER SESSION 2026**

BRANCH: IT **BATCH: REGULAR & KT**

Q.1) Elaborate briefly the simplified 3 layered IoT architecture with neat diagram.

Q.2) Draw and explain the architecture of MQTT with its advantages.

Q.3) Describe the role of NoSQL in IoT Data Analytics Challenges.

Q.4) Describe IoT World Forum (IoTWF) Standardized architecture.

Q.5) Explain the working of Constraint Application Layer Protocol (CoAP).

Q.6) Explain the architecture of LoRaWAN with its major characteristics.

Q.7) Explain the Smart Object with diagram and describe its characteristics.

Q.8) Illustrate the components of IEEE 802.11 architecture with advantages.

Q.9) State various operating modes of Zigbee and its topologies with diagram.

Q.10) Elaborate the need of new network architecture in IoT.

Q.11) Compare and contrast Edge, Fog and Cloud Computing. Explain their usage with an example.

Q.12) Explain the IoT enabled Smart Home automation and Smart Parking system with respect to sensors, Actuators, Framework, Protocol, data analysis and security.

Q.13) Explain effective Data Visualization methods with suitable example.

Q.14) Explain the architecture of RFID with its major applications.

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Q.15) Draw and explain the architecture of Advanced Message Queuing Protocol (AMQP) with advantages.

Q.16) Discuss the functional blocks of IoT architecture, highlighting their roles and interactions. Provide examples to illustrate the importance of each block in the overall functionality of IoT systems.

Q.17) Explain the importance of IoT data analytics.

Q.18) Design the Forest Fire Detection system using IoT sensors.

Q.19) Explain the functional blocks and the characteristics of IoT.

Q.20) Explain the role of HTTP, WebSocket, and MQTT in IoT communication. Compare and contrast these protocols in terms of their characteristics, suitability for different IoT scenarios, and support for real-time data transmission.

Q.21) Elaborate in detail the strategies to organize the data for IoT Analytics.

Q.22) Explain the term in detail: i) Internet of Behaviour (IoB), ii) STOMP, iii) I-IoT

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Q.23) Explain the following access technologies with application area of each: i) IEEE 802.15.4 ii) LTE-A, 6LoWPAN

Q.24) Differentiate between the following:

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Q.23) Explain the following access technologies with application area of each: i) IEEE 802.15.4 ii) LTE-A, 6LoWPAN

Q.24) Differentiate between the following:

i) RFID with Bluetooth
ii) Compare and contrast IoT data analytics and IoT network analytics.
iii) Physical and logical design of IoT

Q.25) Write a short note on:

i) Explain data acquisition in IoT
ii) Data retention strategy
iii) Near Field Communication
iv) Bluetooth Low Energy (BLE) and its application

Q.26) Consider Smark smoke, Smart farming system, Smart Irrigation system, Elaborate its working with block diagram and list down the different types of sensors and actuators required during the deployment scenario.

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TOP 15 QUESTION BANK OF INTERNET OF EVERYTHING

B.E SEM-VII **SCHEME: R-19 C SCHEME** **FOR WINTER SESSION 2026**

BRANCH: IT **BATCH: REGULAR & KT**

Note: ***Refer our complete question bank for better marks and passing.***

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Q.16) Do all the differentiate between, short note and Q.26) from question bank.

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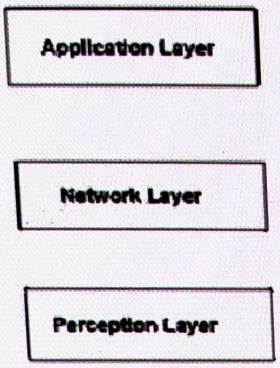
B.E SEM-VII SCHEME: R-19 C SCHEME FOR WINTER SESSION

2026

BRANCH: IT BATCH: REGULAR & KT

Q.1) Elaborate briefly the simplified 3 layered IoT architecture with neat diagram.

Ans: The **three-layer IoT architecture** is a basic framework that explains how an Internet of Things (IoT) system collects data from physical objects, transfers it through a communication network, and uses it to provide useful services. It consists of three layers: the **Perception Layer, Network Layer, and Application Layer**.



3 layer iot architecture

1. Perception Layer (Device Layer)

The Perception Layer is the lowest layer of IoT architecture. It interacts with the physical environment and collects data using sensors and other connected devices.

1) **Data Collection:** Sensors collect information such as temperature, humidity, pressure, light intensity, and motion from the surrounding environment.
2) **Device Identification:** RFID tags, NFC, and other identification technologies help identify physical objects and connected devices.
3) **Actuation:** Actuators perform physical actions based on received commands, such as switching on a light, opening a valve, or controlling a motor.
4) **Environmental Sensing:** Different sensors measure physical conditions and convert them into data that can be processed by an IoT system.

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5) **Data Conversion:** Sensors and embedded devices convert physical measurements into electrical or digital signals for further processing and transmission.

6) **Examples:** Temperature sensors, RFID tags, cameras, motion detectors, Arduino, NodeMCU, and ESP32 devices.

2. Network Layer (Communication Layer)

The Network Layer transfers the data collected by the Perception Layer to processing systems and applications. It also carries control commands back to devices when required.

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1) **Data Transmission:** It transfers collected data between IoT devices, gateways, servers, and cloud platforms.

2) **Connectivity:** It uses communication technologies such as Wi-Fi, Bluetooth, Zigbee, cellular networks, and Ethernet.

3) **Routing and Protocol Handling:** It helps route data to the correct destination using suitable networking protocols.

4) **Gateway Support:** IoT gateways connect devices using different communication technologies and may perform basic data processing before forwarding the data.

5) **Addressing:** Network addressing helps identify devices and endpoints so that data can reach the intended destination.

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6) **Reliable Communication:** Depending on the protocols used, this layer supports appropriate mechanisms for data delivery, error handling, and communication management.

7) **Security:** Communication can be protected using suitable security mechanisms, such as encryption and authentication, to reduce unauthorized access and data interception.

3. Application Layer

The Application Layer is the topmost layer of IoT architecture. It uses the collected and transmitted data to provide services according to the requirements of different users and industries.

1) **Data Processing and Analysis:** It processes IoT data to identify patterns, generate insights, and support decision-

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1) **Data Processing and Analysis:** It processes IoT data to identify patterns, generate insights, and support decision-making.

2) **Monitoring and Control:** It enables users to monitor devices and control their operations through applications, websites, and dashboards.

3) **Industry-Specific Services:** It provides IoT solutions for smart homes, healthcare, agriculture, transportation, and industrial automation.

4) **Data Visualization:** It presents collected information through graphs, charts, reports, and dashboards so that users can understand system conditions.

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5) **Automation and Alerts:** It can generate alerts or trigger automated actions when predefined conditions are met, such as detecting high temperatures or low soil moisture.

6) **Examples:** Smart lighting applications, remote health monitoring systems, smart irrigation, traffic monitoring, and industrial monitoring dashboards.

Working of Three-Layer IoT Architecture

1) The Perception Layer collects data from the physical environment using sensors and connected devices.
2) The Network Layer transfers the collected data to the required processing systems through communication networks.
3) The Application Layer processes the data and provides useful information or services to users.
4) Based on the analyzed data, the application may generate alerts or send control instructions to the relevant devices.
5) When necessary, control commands travel back through the network to actuators, which perform the required physical actions.

Thus, the simplified three-layer IoT architecture organizes an IoT system into sensing, communication, and application layers, enabling physical devices to exchange data and provide intelligent services.

Q.2) Draw and explain the architecture of MQTT with its advantages.

Ans: MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol used for communication between IoT devices over a network. It follows the **publish-subscribe communication model**, in which publishers send messages to an MQTT broker, and subscribers receive messages from the broker according to their subscribed topics. It is suitable for IoT applications with limited bandwidth, low power, and unreliable network connections.

MQTT Architecture

The main components of MQTT architecture are publishers, an MQTT broker, and subscribers.

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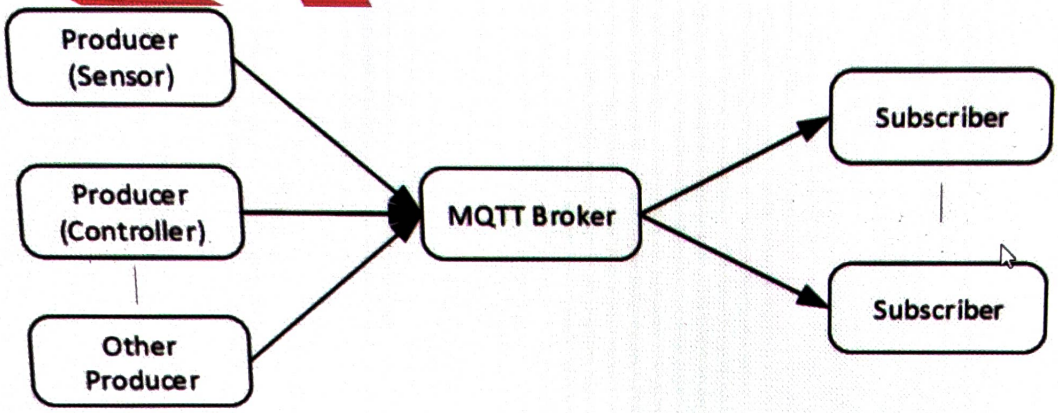

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1. Publisher (Producer): A publisher is an IoT device or application that generates and sends data to the MQTT broker.

1) **Data Generation:** It collects or generates information such as temperature, humidity, motion, or device status.
2) **Message Publishing:** It publishes messages to the broker under a specific topic, such as home/temperature.
3) **Examples:** Sensors, controllers, smart meters, and other IoT devices can act as publishers.

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2. MQTT Broker: The MQTT broker is the central component that receives messages from publishers and forwards them to the appropriate subscribers.

1) **Message Reception:** It receives messages published by different devices.
2) **Topic Management:** It uses topics to organize messages and determine which subscribers should receive them.
3) **Message Distribution:** It forwards published messages to all subscribers whose topic subscriptions match the message topic.
4) **Client Management:** It manages client connections and subscriptions and handles communication between publishers and subscribers.
5) **Quality of Service (QoS):** It supports different delivery levels to provide different degrees of message-delivery assurance.
6) **Security:** Depending on its configuration, the broker can support authentication, access control, and encrypted communication.

3. Subscriber (Receiver): A subscriber is an IoT device or application that receives messages from the broker after subscribing to relevant topics.

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1) **Topic Subscription:** A subscriber registers interest in a particular topic, such as home/temperature.
2) **Message Reception:** The broker forwards matching messages to the subscriber when they are published.
3) **Data Utilization:** The received data can be displayed, stored, analyzed, or used to trigger actions.
4) **Examples:** Mobile applications, monitoring dashboards, cloud applications, and smart home controllers can act as subscribers.

4. Topic-Based Communication: MQTT uses topics to organize and route messages between publishers and subscribers.

1) A topic is a hierarchical text name, such as factory/machine1/temperature.
2) Publishers send messages to topics, while subscribers subscribe to the topics they want to receive.
3) Publishers and subscribers do not need to know each other directly; the broker manages message distribution.

Working of MQTT Architecture

1) The publisher collects data from a sensor or another device and creates a message.
2) The publisher sends the message to the MQTT broker under a specific topic.
3) The broker checks the topic against the subscriptions registered by connected subscribers.

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3) The broker checks the topic against the subscriptions registered by connected subscribers.
4) The broker forwards the message to the matching subscribers.
5) Subscribers use the received data for monitoring, analysis, storage, or automated control.

Example: In a smart home, a temperature sensor publishes readings to the topic home/temperature. The MQTT broker forwards these readings to a mobile application subscribed to that topic, allowing the user to monitor the room temperature.

Advantages of MQTT

1) **Lightweight Protocol:** MQTT has a small protocol overhead, making it suitable for resource-constrained IoT devices.
2) **Low Bandwidth Usage:** It minimizes communication overhead and is useful for networks with limited bandwidth.
3) **Efficient Communication:** Its publish-subscribe model allows messages to be distributed without requiring publishers to communicate directly with every subscriber.
4) **Scalability:** A broker can manage communication among many connected publishers and subscribers, subject to its capacity and configuration.
5) **Reliable Message Delivery:** MQTT supports three QoS levels: at most once (QoS 0), at least once (QoS 1), and exactly once (QoS 2).
6) **Loose Coupling:** Publishers and subscribers do not need to know each other's identities; they communicate through the broker and topics.
7) **Support for Unreliable Networks:** MQTT includes features such as persistent sessions and message retention that can help applications handle intermittent connectivity when configured appropriately.
8) **Security Support:** MQTT deployments can use authentication, access controls, and TLS encryption to protect communication.

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9) **Bidirectional Communication:** Devices can publish data and subscribe to commands, supporting remote monitoring and control.

10) **Wide Range of Applications:** MQTT is used in smart homes, industrial monitoring, healthcare, agriculture, smart cities, and remote telemetry.

Thus, MQTT provides lightweight and efficient message-based communication through publishers, a central broker, and subscribers, making it well suited for IoT systems that require flexible and scalable data exchange.

Q.3) Describe the role of NoSQL in IoT Data Analytics Challenges.

Ans: The **Internet of Things (IoT)** generates large amounts of data through sensors, smart devices, machines, and connected systems. Analyzing this data is challenging because it is produced continuously, arrives at high speed, and exists in different formats. **NoSQL databases** help address these challenges by providing flexible data models, scalability, and efficient data storage and retrieval for many IoT applications.

Challenges in IoT Data Analytics

1) **Large Volume of Data:** IoT devices generate huge amounts of data continuously. Traditional databases may require careful scaling and optimization to handle such workloads efficiently.

2) **High Velocity of Data:** Sensors frequently transmit real-time readings, requiring databases to support rapid data ingestion and processing.

3) **Variety of Data:** IoT data includes structured, semi-structured, and unstructured formats, such as sensor readings, JSON messages, images, and device logs.

4) **Scalability:** As the number of connected devices increases, the database must handle growing data volumes and increasing numbers of requests.

5) **Real-Time Analytics:** Applications such as industrial monitoring and traffic management need timely analysis to identify abnormal conditions and support quick decisions.

6) **Data Integration:** Data collected from different devices, communication protocols, and platforms must be combined for meaningful analysis.

7) **Data Storage and Retention:** IoT systems need suitable methods to store historical readings, manage data retention periods, and control storage costs.

8) **Data Quality and Reliability:** Sensor readings may contain missing values, duplicate records, noise, or incorrect measurements, which can affect analytical results.

Role of NoSQL in IoT Data Analytics

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Role of NoSQL in IoT Data Analytics

NoSQL databases are non-relational databases designed to support flexible data models and different data-management requirements. Their role in IoT analytics includes the following:

1) **Flexible Data Models:** NoSQL databases can store data without requiring every record to follow a fixed relational table structure. This is useful when different IoT devices produce different data fields.
2) **High-Speed Data Ingestion:** Many NoSQL systems are designed to handle large numbers of write operations, making them suitable for continuously arriving sensor data.
3) **Horizontal Scalability:** Distributed NoSQL databases can scale across multiple servers by distributing data and workloads, helping accommodate growing numbers of IoT devices.
4) **Handling Diverse Data:** Document databases can store JSON-like records, key-value databases can store device states, and wide-column databases can support large distributed datasets.
5) **Distributed Storage:** Data can be distributed across multiple nodes to improve capacity and, depending on the database configuration, availability and fault tolerance.
6) **Real-Time Data Access:** NoSQL databases can provide efficient retrieval of recent device readings, status information, and events when the data model and queries are designed appropriately.
7) **Support for Analytics Pipelines:** NoSQL databases can serve as a data-storage layer for applications that process IoT data, detect patterns, and generate reports or alerts.
8) **Cost-Effective Expansion:** Open-source NoSQL solutions and commodity-server deployments can help organizations expand storage capacity without relying exclusively on large centralized database servers.
9) **Device and Metadata Management:** They can store device identifiers, configuration details, operating status, and sensor metadata alongside or separately from time-series measurements.
10) **Integration with Cloud Platforms:** NoSQL databases are available in cloud-managed services and can be integrated into cloud-based IoT data-processing systems.

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Types of NoSQL Databases Used in IoT

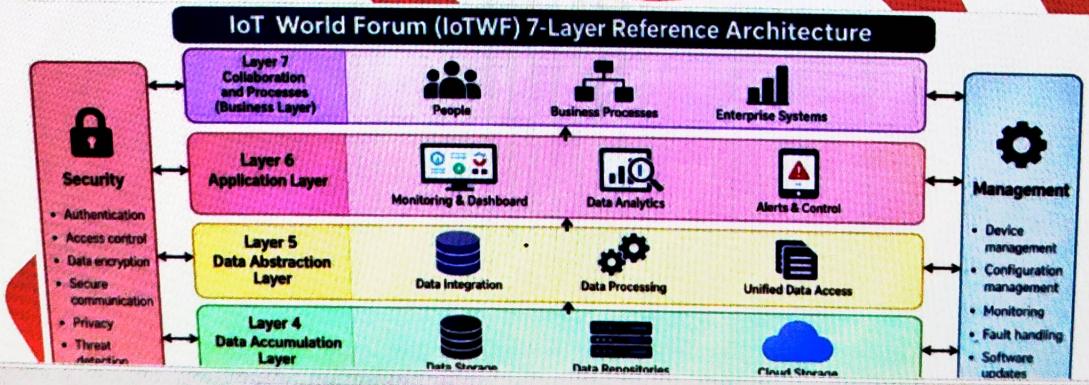
1) **Key-Value Databases:** Store data as key-value pairs and are useful for device states, configurations, and rapidly accessed information. Example: Redis.
2) **Document Databases:** Store flexible documents, often in JSON-like formats, making them suitable for sensor messages and device metadata. Example: MongoDB.
3) **Wide-Column Databases:** Organize data into column families and are designed for distributed storage of large datasets. Example: Apache Cassandra.
4) **Graph Databases:** Represent information as nodes and relationships, which can help analyze connections among devices, users, and components. Example: Neo4j.

Example of NoSQL in IoT Analytics: Consider a smart agriculture system in which sensors measure soil moisture, temperature, and humidity. The sensors continuously send readings to a data platform, where a NoSQL database stores the incoming records. An analytics application examines the readings and identifies when soil moisture falls below a required level. The system can then generate an alert or initiate irrigation through a connected controller.

Thus, NoSQL helps address major IoT data analytics challenges by supporting flexible data storage, high-volume data ingestion, and horizontal scalability. However, the appropriate database must be selected according to the application's data structure, query requirements, consistency needs, and analytical workload.

Q.4) Describe IoT World Forum (IoTWF) Standardized architecture.

Ans: The **IoT World Forum (IoTWF) standardized architecture** is a seven-layer reference model developed to organize the major functions of an Internet of Things system. It provides a structured framework for connecting physical devices, processing data, transmitting information, and delivering business applications. The model consists of **seven layers**, with security and management functions extending across all layers.



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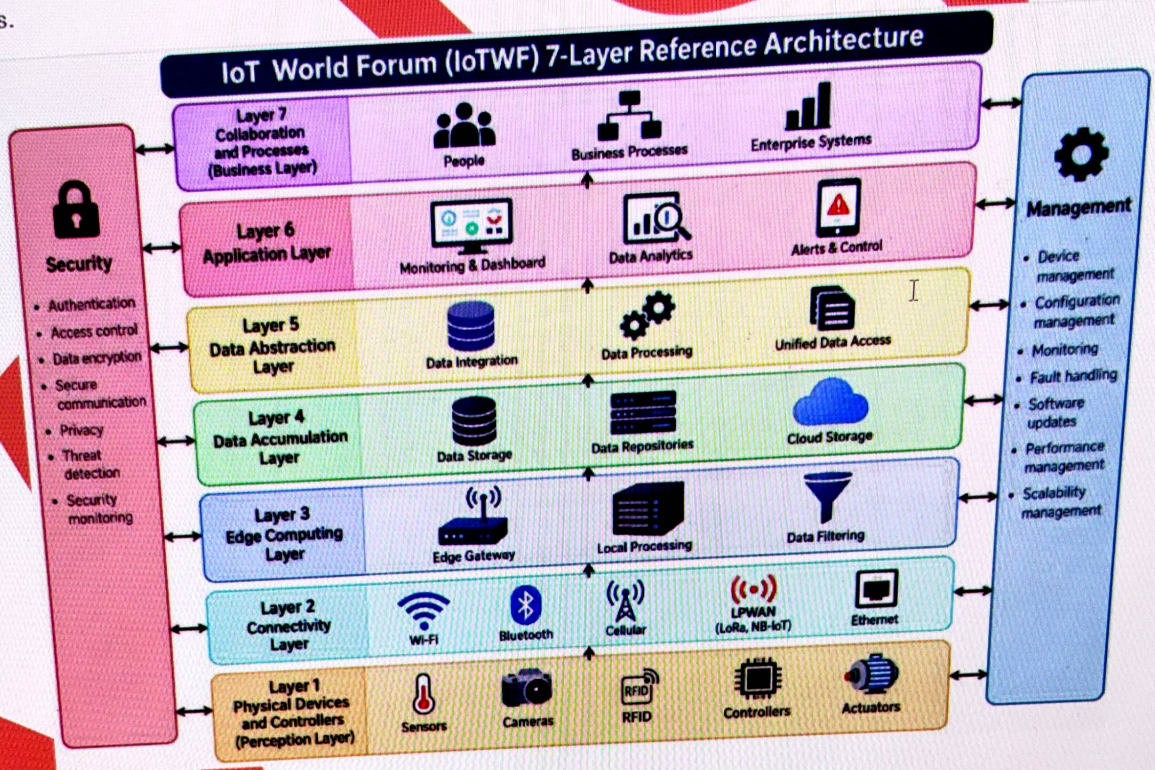

Layers of IoTWF Standardized Architecture

1. Physical Devices and Controllers Layer

This is the lowest layer of the architecture. It includes physical objects, sensors, actuators, machines, and controllers that interact with the real world.

1) **Data Collection:** Sensors collect physical measurements such as temperature, humidity, pressure, and motion.
2) **Physical Actions:** Actuators perform operations such as opening valves, controlling motors, and switching lights.
3) **Device Control:** Embedded controllers manage the operation of connected devices.

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3) **Device Control:** Embedded controllers manage the operation of connected devices.

4) **Examples:** Smart meters, industrial sensors, RFID readers, cameras, and microcontrollers.

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2. Connectivity Layer

The Connectivity Layer provides communication between physical devices, gateways, and other system components. It transfers data from devices to higher layers and carries control commands in the reverse direction.

1) **Communication:** It enables devices to exchange information through wired and wireless networks.

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2) **Network Technologies:** It may use Ethernet, Wi-Fi, Bluetooth, cellular networks, and LPWAN technologies.

3) **Data Transport:** It supports the transfer of device data through appropriate communication protocols.

4) **Network Reliability:** It helps maintain communication between devices and the IoT infrastructure.

3. Edge Computing Layer

The Edge Computing Layer performs processing close to the physical devices rather than sending all raw data to a centralized cloud or data center.

1) **Local Processing:** It processes data near the source, reducing the need to transmit all raw readings.

2) **Filtering:** It removes unnecessary or irrelevant data before forwarding it to higher layers.

3) **Rapid Response:** It supports time-sensitive decisions, such as stopping a machine when an abnormal condition is detected.

4) **Protocol Translation:** Edge gateways may translate between different device communication protocols.

5) **Reduced Network Load:** Local processing can reduce bandwidth consumption and communication delays.

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4. Data Accumulation Layer

The Data Accumulation Layer converts data in motion into data at rest. It stores incoming data so that it can be accessed and processed later.

1) **Data Storage:** It stores incoming IoT data in databases, storage systems, or data repositories.
2) **Data Buffering:** It temporarily retains data when downstream processing systems cannot process it immediately.
3) **Data Organization:** It organizes stored information for subsequent retrieval and analysis.
4) **Separation of Processing:** It separates continuous data collection from later data-processing activities.

5. Data Abstraction Layer

The Data Abstraction Layer makes stored IoT data easier to access, integrate, and use. It provides a consistent view of data collected from different devices and systems.

1) **Data Integration:** It combines data from multiple sources into a usable representation.
2) **Data Organization:** It structures and prepares information for analytical applications.
3) **Data Access:** It provides appropriate interfaces for higher layers to retrieve required data.
4) **Data Filtering and Selection:** It helps make relevant information available to applications without exposing unnecessary underlying details.
5) **Data Preparation:** It may support cleaning, transformation, and normalization before data is used for analysis.

6. Application Layer

The Application Layer provides software applications that use IoT data for monitoring, analysis, visualization, and control.

1) **Monitoring:** It allows users to observe the status and performance of connected devices.
2) **Data Visualization:** It presents information through dashboards, graphs, charts, and reports.
3) **Analytics:** It supports the analysis of IoT data to identify patterns, trends, and abnormal conditions.
4) **Alerts:** It generates notifications when predefined conditions or thresholds are reached.
5) **Examples:** Smart home applications, healthcare monitoring systems, industrial dashboards, and smart agriculture applications.

7. Collaboration and Processes Layer

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7. Collaboration and Processes Layer

This is the highest layer of the IoTWF architecture. It connects IoT-generated information with people, business processes, and organizational decision-making.

1) **Decision-Making:** It uses information from IoT applications to support operational and business decisions.
2) **Collaboration:** It enables different teams and departments to share IoT insights.
3) **Business Process Integration:** It integrates IoT information with workflows and enterprise systems.
4) **Process Automation:** It can initiate business or operational actions based on IoT events and analytical results.
5) **Business Value:** It helps organizations improve efficiency, reduce downtime, and optimize resource utilization.

Cross-Layer Functions: Security and Management

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Security and management are not separate numbered layers in the seven-layer model. They are cross-layer functions that apply throughout the architecture.

1) **Security:** It includes device authentication, access control, data protection, secure communication, and monitoring for threats.
2) **Management:** It includes device configuration, system monitoring, software updates, fault handling, and performance management.

Working of IoTWF Architecture

1) Physical devices and controllers collect data from the environment.
2) The Connectivity Layer transfers the data through suitable communication networks.
3) The Edge Computing Layer processes and filters data near its source.
4) The Data Accumulation Layer stores incoming data for later use.

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4) The Data Accumulation Layer stores incoming data for later use.

5) The Data Abstraction Layer organizes and prepares stored data for applications.

6) The Application Layer provides monitoring, analytics, and visualization services.

7) The Collaboration and Processes Layer uses the resulting information to support decisions and business activities.

Thus, the IoTWF standardized architecture divides an IoT system into seven functional layers, from physical devices to business processes, while security and management operate across the entire architecture.

Q.5) Explain the working of Constraint Application Layer Protocol (CoAP).

Ans: CoAP (Constrained Application Protocol) is a lightweight application-layer protocol designed for communication between constrained IoT devices and systems with limited memory, processing power, and bandwidth. It follows a **request-response model**, similar to HTTP, but is optimized for resource-constrained devices. CoAP commonly uses UDP for communication and supports methods such as GET, POST, PUT, and DELETE.

Working of CoAP:

1) **Client-Server Communication:** CoAP uses a client-server model in which a client sends a request to a server, and the server returns a response. The client may be a monitoring application, while the server may be an IoT device providing sensor data.
2) **Resource Identification:**
   - CoAP represents available device information or functions as resources.
   - Each resource is identified using a URI, such as coap://sensor.local/temperature.
   - Clients access these resources to retrieve sensor readings or perform operations.
3) **Request Methods:** CoAP supports four main request methods are as follows,
   - **GET:** Retrieves information from a resource, such as the current temperature.
   - **POST:** Sends data to a server or requests it to perform an operation.
   - **PUT:** Creates or updates the state of a resource.
   - **DELETE:** Removes a specified resource, where supported.
4) **Message Transmission**
   - The client sends a CoAP request to the server, commonly over UDP.
   - The request contains information such as the method, resource address, message identifier, and optional payload.

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- The request contains information such as the method, resource address, message identifier, and optional payload.
- The server receives the request and processes the requested operation.
- The server sends a response containing a status code and, when required, the requested data.

5) **Message Reliability:** CoAP supports different message types to handle communication reliability.
   - **Confirmable (CON):** Requires an acknowledgement from the receiver. If an acknowledgement does not arrive, retransmission may occur.
   - **Non-confirmable (NON):** Does not require an acknowledgement and is suitable when occasional message loss is acceptable.
   - **Acknowledgement (ACK):** Confirms the receipt of a confirmable message.
   - **Reset (RST):** Indicates that a message was received but could not be processed in the current context.

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6) **Response Codes:** CoAP uses response codes to indicate the result of a request, such as successful processing, a missing resource, or a server error. These codes help the client understand whether the requested operation succeeded.
7) **Resource Observation:** CoAP supports an Observe mechanism that allows a client to register interest in a resource. The server can then send updated values when the resource changes, reducing the need for repeated polling.

Example of CoAP Working: Consider an IoT temperature sensor that provides readings through a CoAP server.

1) A client sends a GET request to coap://sensor.local/temperature.
2) The CoAP server reads the current temperature from the sensor.
3) The server returns a response containing the temperature value, for example, 28°C.
4) The client displays the received reading in a monitoring application.

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- The request contains information such as the method, resource address, message identifier, and optional payload.
- The server receives the request and processes the requested operation.
- The server sends a response containing a status code and, when required, the requested data.

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4) The client displays the received reading in a monitoring application.

Advantages of CoAP

1) **Lightweight Communication:** Its compact message format reduces protocol overhead.
2) **Low Resource Requirements:** It is suitable for devices with limited memory and processing power.
3) **Efficient Network Usage:** Its design is appropriate for low-bandwidth and constrained networks.
4) **Multicast Support:** CoAP can support multicast communication, allowing requests to be sent to multiple devices where the network permits it.
5) **Resource Observation:** It supports notification of resource changes without requiring continuous repeated requests.
6) **Security Support:** CoAP deployments can use suitable security mechanisms, such as DTLS for UDP-based communication or OSCORE for protecting CoAP messages.

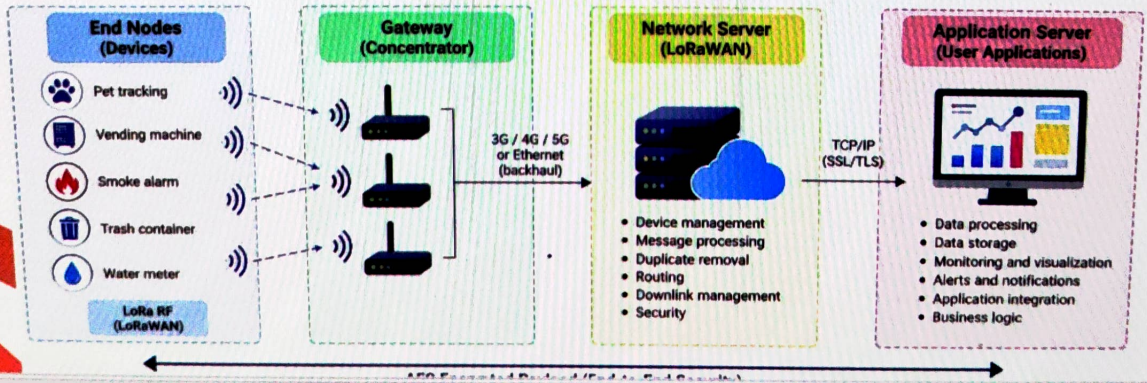
Thus, CoAP enables efficient request-response communication between constrained IoT devices by providing lightweight messaging, resource identification, reliability options, and support for monitoring changing device data.

Q.6) Explain the architecture of LoRaWAN with its major characteristics.

Ans: LoRaWAN (Long Range Wide Area Network) is a Low Power Wide Area Network (LPWAN) communication protocol designed to connect IoT devices over long distances while consuming very little power. It uses **LoRa (Long Range)** radio technology for wireless communication and defines how end devices communicate with gateways, network servers, and application servers.

Architecture of LoRaWAN

The LoRaWAN architecture consists of four major components: **End Nodes, Gateways, Network Server, and Application Server.**



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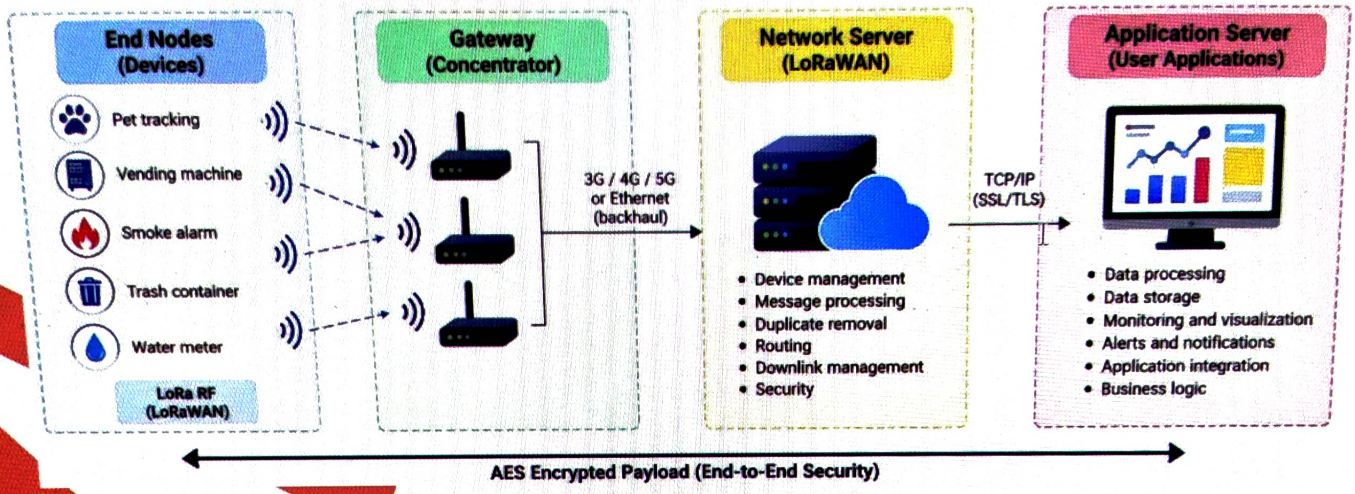

1. End Nodes (End Devices)

End nodes are IoT devices equipped with sensors or controllers that collect data from the physical environment.

1) **Data Collection:** They collect information such as temperature, water levels, smoke, and device status.
2) **Wireless Transmission:** They transmit collected data using LoRa radio technology.
3) **Low-Power Operation:** They are designed to operate with low energy consumption, allowing battery-powered devices to function for extended periods.
4) **Examples:** Water meters, smoke alarms, pet-tracking devices, vending machines, and smart waste containers.

2. Gateway (Concentrator)

A gateway receives LoRa radio messages from nearby end devices and forwards them to the network server through an IP-based backhaul connection.

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1) **Message Reception:** It receives wireless messages transmitted by multiple LoRaWAN end devices.
2) **Data Forwarding:** It forwards received messages to the network server using connections such as Ethernet or cellular networks.
3) **Two-Way Communication:** It can transmit downlink messages from the network server to end devices when required.
4) **Protocol Bridging:** It connects the LoRa radio network to the IP-based network infrastructure.
5) **Important Point:** Gateways generally relay messages; the network server performs the main network-level processing.

3. Network Server

The network server manages communication between end devices, gateways, and application servers.

1) **Message Processing:** It processes incoming LoRaWAN messages received through gateways.
2) **Duplicate Removal:** Since multiple gateways may receive the same transmission, the network server can identify and eliminate duplicate messages.
3) **Device Management:** It manages device sessions and network communication parameters.
4) **Message Routing:** It forwards application data to the appropriate application server.
5) **Downlink Management:** It coordinates the transmission of downlink messages through suitable gateways.
6) **Security:** It supports network-level security functions, including message integrity checks and network session management.

4. Application Server

The application server processes the information received from end devices and provides services to users or applications.

1) **Data Processing:** It interprets the application data sent by IoT devices.
2) **Data Storage:** It can store sensor readings for historical analysis and reporting.
3) **Monitoring and Visualization:** It displays data through dashboards, reports, and monitoring applications.
4) **Alerts and Automation:** It can generate alerts or initiate actions when specified conditions are detected.
5) **Examples:** Smart agriculture platforms, water-monitoring systems, smart-city applications, and industrial monitoring systems.

Working of LoRaWAN Architecture

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Working of LoRaWAN Architecture

1) The end node collects data using sensors, such as a water-level sensor measuring the level in a tank.
2) The end node transmits the data wirelessly to one or more nearby LoRaWAN gateways.
3) The gateway forwards the received message to the network server through an IP-based connection.
4) The network server processes the message, handles duplicates, and routes the application data to the appropriate application server.
5) The application server processes the data and displays the result or generates an alert for the user.
6) If a response or control command is required, it can travel back through the network server and gateway to the end device, subject to LoRaWAN communication and device-class constraints.

Major Characteristics of LoRaWAN

1) **Long Communication Range:** It supports communication over long distances. The actual range depends on terrain, buildings, antenna placement, and radio conditions.
2) **Low Power Consumption:** Its design allows battery-powered IoT devices to operate for long periods with suitable transmission settings and usage patterns.
3) **Wide Area Coverage:** A gateway can communicate with many end devices distributed across a large area.
4) **Low Data Rate:** LoRaWAN is designed for small data messages rather than high-bandwidth applications such as video streaming.
5) **Unlicensed Radio Spectrum:** LoRaWAN commonly operates in region-specific unlicensed frequency bands, subject to local regulations.
6) **Star-of-Stars Topology:** End devices communicate with gateways, which relay messages to a central network server.
7) **Bidirectional Communication:** It supports uplink messages from devices and downlink messages to devices, although downlink capacity is limited by radio and regional constraints.

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8) **Security:** LoRaWAN provides security mechanisms based on cryptographic keys, including network and application-level protection.

9) **Scalability:** Its architecture can support large numbers of devices when the network is properly designed and configured.

10) **Application Versatility:** It is used in smart agriculture, smart metering, environmental monitoring, waste management, and smart-city systems.

11) **Limited Payload and Throughput:** Message sizes and transmission rates are constrained, making it unsuitable for applications that require continuous high-volume data transfer.

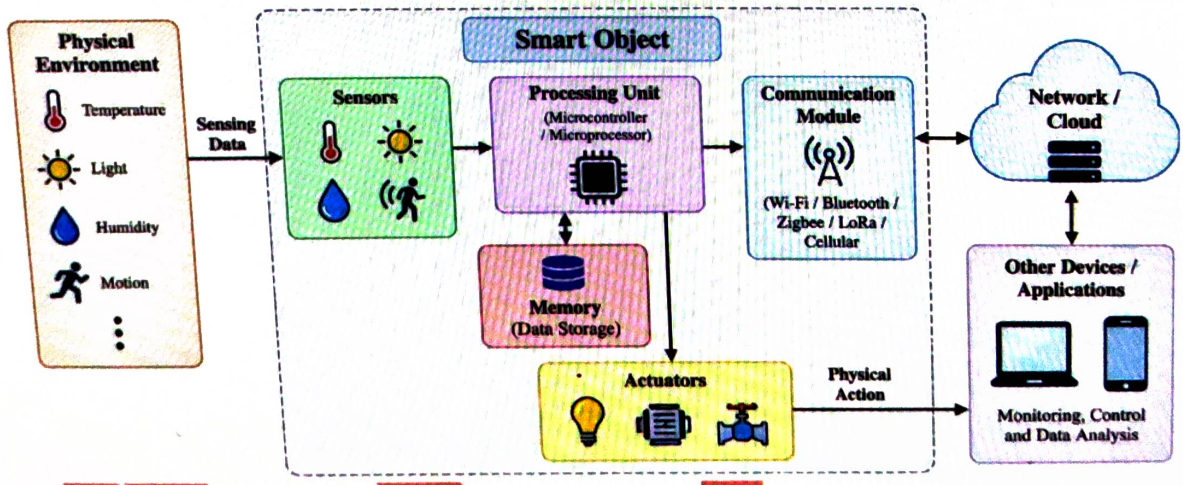
12) **Asynchronous Communication:** Many LoRaWAN devices transmit data intermittently rather than maintaining continuous connections, which helps reduce power consumption.

Thus, LoRaWAN provides long-range, low-power wireless connectivity for IoT devices through end nodes, gateways, a network server, and an application server, making it suitable for applications that transmit small amounts of data over wide areas.

Q.7) Explain the Smart Object with diagram and describe its characteristics.

Ans: A **Smart Object** is a physical object equipped with sensing, processing, communication, and sometimes actuation capabilities. It can collect data from its environment, process the data, and communicate with other devices or systems through a network. Smart objects are important components of IoT because they enable physical objects to perform intelligent tasks and provide automated services.

Architecture of a Smart Object



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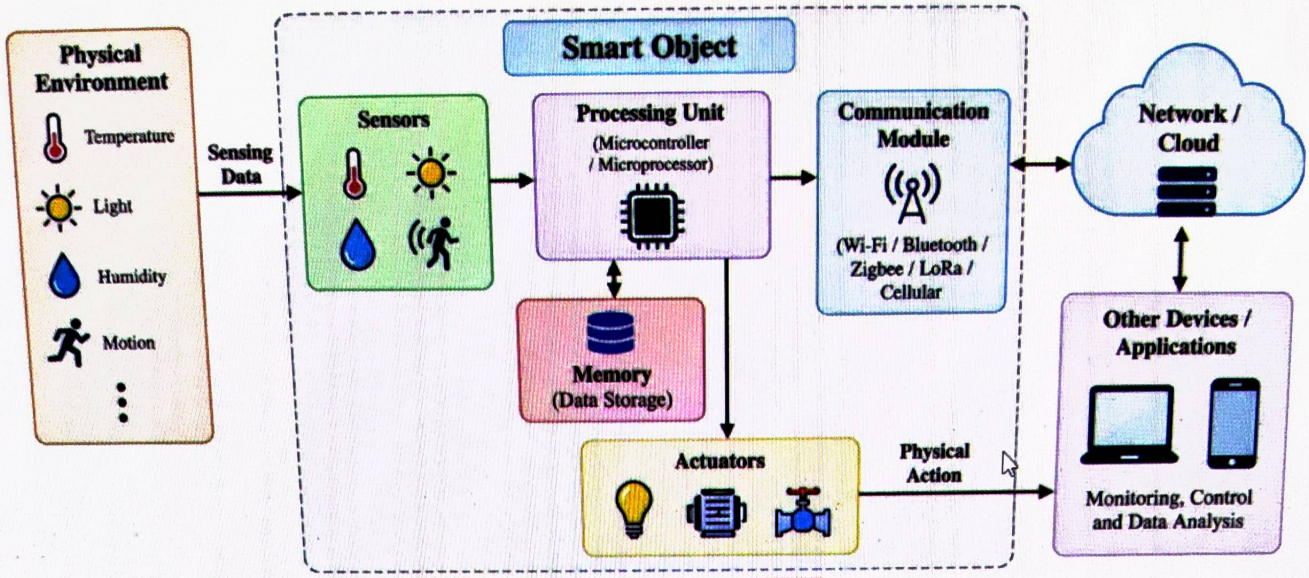
Architecture of a Smart Object



The main components of a smart object are sensors, a processing unit, memory, a communication module, and actuators.

1) **Sensors:** Sensors collect data from the physical environment, such as temperature, humidity, light, and motion.
2) **Processing Unit:** A microcontroller or microprocessor processes the collected data and makes decisions based on predefined instructions.
3) **Memory (Data Storage):** Memory stores sensor readings, program instructions, configuration settings, and device status information.
4) **Communication Module:** It enables the smart object to exchange data with other devices, gateways, or cloud platforms using technologies such as Wi-Fi, Bluetooth, Zigbee, LoRa, or cellular networks.
5) **Actuators:** Actuators perform physical actions based on processed data or received commands, such as switching on a light, opening a valve, or controlling a motor.
6) **Network or Cloud Connectivity:** The smart object may connect to a network or cloud platform for remote monitoring, data storage, analysis, and control.

Characteristics of a Smart Object

1) **Sensing Capability:** A smart object can collect information from its surroundings using sensors. For example, a temperature sensor measures the surrounding temperature.

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temperature sensor measures the surrounding temperature.

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2) **Processing Capability:** It has a processing unit that processes collected data and can make local decisions based on programmed logic.

3) **Communication Capability:** It can exchange information with other devices or systems through wired or wireless communication technologies.

4) **Actuation Capability:** Some smart objects can perform physical actions using actuators. For example, a smart thermostat can control a heating or cooling system.

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5) **Unique Identification:** A smart object can be identified individually using an identifier, device ID, or suitable network address.

6) **Data Storage:** It may contain memory to store sensor readings, configuration parameters, and device status information.

7) **Autonomous Operation:** It can perform certain tasks automatically without continuous human intervention, depending on its capabilities and configuration.

8) **Connectivity and Interoperability:** It can communicate and work with compatible devices, gateways, and applications within an IoT system.

9) **Energy Efficiency:** Many smart objects are designed to consume less power, especially when operating on batteries or in remote locations.

10) **Context Awareness:** By using sensor data, a smart object can respond to changes in its environment, such as switching on a light when movement is detected.

Example of a Smart Object: A smart irrigation system uses soil-moisture sensors to measure the moisture level in the soil. A processing unit analyzes the readings and activates a water pump through an actuator when the soil becomes too dry. The system may also transmit the readings to a mobile application for remote monitoring.

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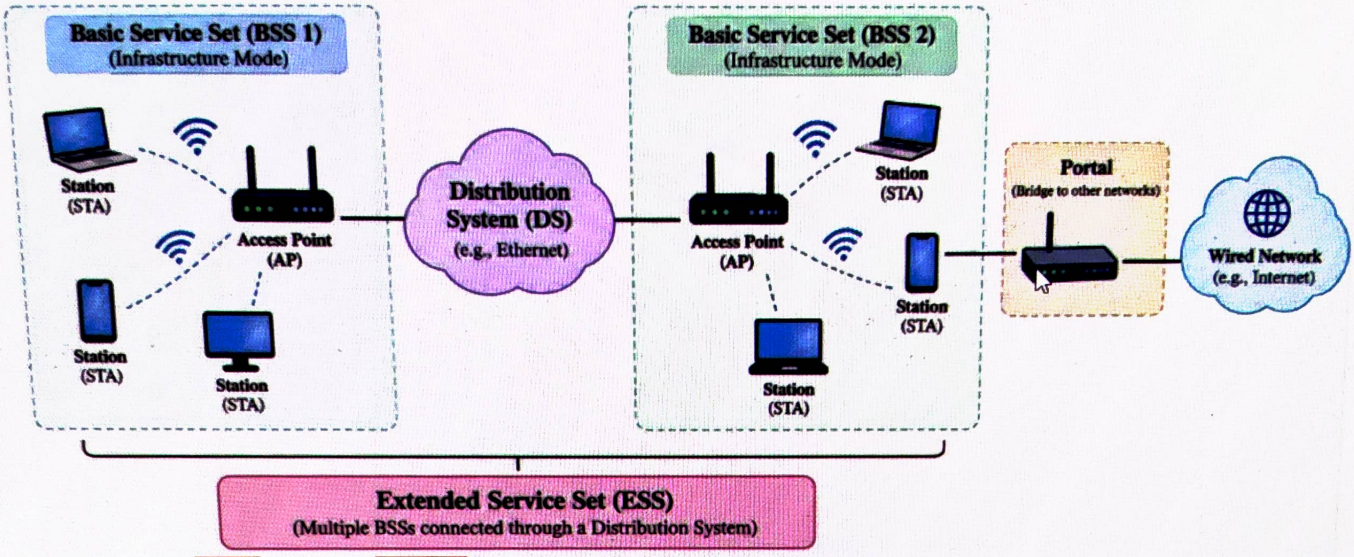
Q.8) Illustrate the components of IEEE 802.11 architecture with advantages.

Ans: IEEE 802.11 is a family of standards developed by the Institute of Electrical and Electronics Engineers (IEEE) for Wireless Local Area Networks (WLANs). It defines the rules for wireless communication between devices using Wi-Fi. Its architecture consists of stations, access points, Basic Service Sets, a Distribution System, Extended Service Sets, and portals.

Components of IEEE 802.11 Architecture:



1. Station (STA): A station is any device equipped with an IEEE 802.11 wireless network interface that can communicate over a Wi-Fi network.

- It can transmit and receive wireless data.
- Stations may include laptops, smartphones, tablets, and other wireless devices.
- A station communicates directly with another station in an independent network or through an access point in an infrastructure network.

2. Access Point (AP): An access point is a central device that connects wireless stations to a network.

- It allows multiple wireless devices to communicate within its coverage area.
- It connects wireless stations to the Distribution System.
- It helps coordinate wireless communication and network access.

3. Basic Service Set (BSS): A Basic Service Set is a group of wireless stations that communicate within a single basic wireless

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3. Basic Service Set (BSS): A Basic Service Set is a group of wireless stations that communicate within a single basic wireless network.

- In an infrastructure BSS, stations communicate through an access point.
- It represents a basic wireless coverage area.
- Each BSS is identified by a Basic Service Set Identifier (BSSID).

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4. Distribution System (DS): The Distribution System connects multiple access points and allows communication between different BSSs.

- It forwards data between access points.
- It commonly uses Ethernet or another suitable network infrastructure.
- It helps extend wireless network coverage across multiple areas.

5. Extended Service Set (ESS): An Extended Service Set consists of two or more infrastructure BSSs connected through a Distribution System.

- It expands wireless coverage beyond a single BSS.
- It allows users to move between access points while maintaining network connectivity, subject to network configuration.
- It is commonly used in colleges, offices, hospitals, and large buildings.

6. Portal: A portal connects an IEEE 802.11 WLAN to another type of network.

- It provides connectivity between a wireless network and networks such as a wired LAN.
- It enables wireless devices to access external network resources.

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- It enables wireless devices to access external network resources.
- A router or gateway may provide this functionality.

Working of IEEE 802.11 Architecture

1) Wireless stations connect to an access point using Wi-Fi.
2) The access point forwards data between wireless stations and the Distribution System.
3) The Distribution System connects different BSSs and transfers data between access points.
4) Multiple BSSs connected through the Distribution System form an ESS, extending network coverage.
5) A portal connects the WLAN to other networks, such as a wired LAN or the Internet.

Advantages of IEEE 802.11 Architecture

1) **Mobility and Flexibility:** Users can access network services while moving within the wireless coverage area without requiring a physical cable connection.
2) **Easy Installation:** Wireless networks reduce the need for extensive cabling, making installation simpler in many environments.
3) **Scalability:** Additional access points can be installed to expand coverage and support more users.
4) **Cost-Effective Deployment:** It can reduce cabling and installation costs, particularly in buildings where wired connections are difficult to install.
5) **Interoperability:** Standardized Wi-Fi technologies allow compatible devices from different manufacturers to communicate.
6) **Roaming Support:** An ESS can allow users to move between access points while maintaining network connectivity when roaming is properly configured.
7) **Wide Range of Applications:** IEEE 802.11 is used in homes, educational institutions, offices, hospitals, public hotspots, and industrial environments.

Thus, IEEE 802.11 provides a standardized architecture for wireless networking through stations, access points, BSSs, a Distribution System, ESSs, and portals, enabling flexible, scalable, and convenient Wi-Fi communication.

Q.9) State various operating modes of Zigbee and its topologies with diagram.

Ans: Zigbee is a low-power, low-data-rate wireless communication technology based on the IEEE 802.15.4 standard. It is widely used in IoT applications such as home automation, smart lighting, industrial monitoring, and wireless sensor networks. Zigbee defines three main device roles: Coordinator, Router, and End Device. It also supports three major network topologies: Star, Tree, and Mesh.

Operating Modes (Device Roles) of Zigbee

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Operating Modes (Device Roles) of Zigbee

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Zigbee networks consist of three types of devices, each performing a specific function.

1. **Coordinator (ZC)**: The Zigbee Coordinator is responsible for establishing and managing the network.
   1) **Network Formation:** It initializes and establishes the Zigbee network.
   2) **Channel Selection:** It selects a suitable radio channel and Personal Area Network (PAN) identifier.
   3) **Device Joining:** It permits and manages the joining of devices according to network security and configuration settings.
   4) **Network Management:** It maintains important network information and coordinates network operations.

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   5) **Communication:** It can communicate with routers and end devices within the network.
   6) **Uniqueness:** A Zigbee network has only one coordinator.
2. **Router (ZR)**: A Zigbee Router forwards messages between devices and helps extend network coverage.
   1) **Data Forwarding:** It receives and forwards messages between devices.
   2) **Network Extension:** It extends the coverage area by allowing communication through additional routers.
   3) **Device Joining:** It can allow other routers and end devices to join the network through it.
   4) **Multi-Hop Communication:** It enables data to travel through intermediate routers when the destination is outside direct radio range.
   5) **Network Availability:** It generally remains available to forward messages for its child devices.
   6) **Power Requirement:** It usually consumes more power than a sleeping end device because it must remain available for network communication.
3. **End Device (ZED)**: A Zigbee End Device performs specific sensing or control tasks and communicates through its parent

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3. End Device (ZED): A Zigbee End Device performs specific sensing or control tasks and communicates through its parent coordinator or router.

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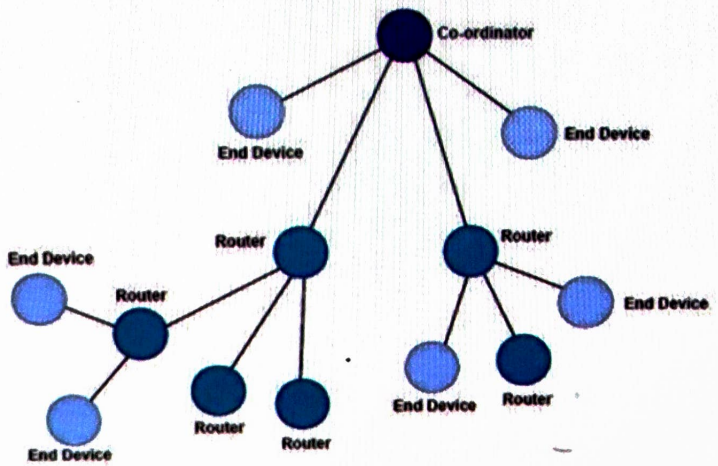
1) **Specific Function:** It performs tasks such as measuring temperature, detecting motion, or controlling a light.
2) **Parent Communication:** It sends and receives data through its parent coordinator or router.
3) **No Routing:** It does not forward messages for other network devices.
4) **Low Power Consumption:** It is designed to operate with low energy consumption and is suitable for battery-powered devices.
5) **Sleep Mode:** It can enter sleep mode between communication periods to conserve battery power, depending on its configuration.
6) **Applications:** Examples include temperature sensors, motion detectors, wireless switches, and simple control devices.

Zigbee Network Topologies

A network topology describes how devices are connected and how data travels between them. Zigbee supports three main topologies: Star, Tree, and Mesh.

1. Star Topology: In Star topology, all end devices communicate through a central coordinator. The end devices do not forward data for one another.

Diagram:



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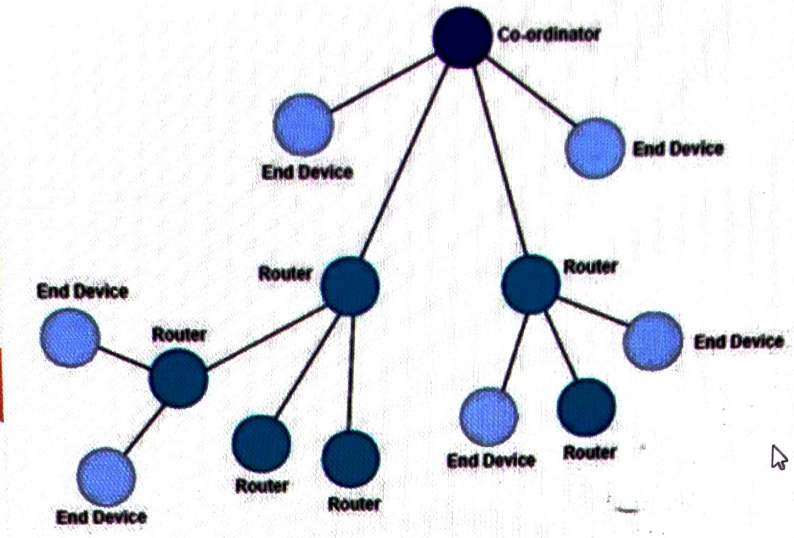

Characteristics:

1) **Centralized Structure:** The coordinator is the central device, and other devices communicate through it.
2) **Simple Design:** The network is easy to install, configure, and maintain.
3) **Direct Communication:** End devices communicate with the coordinator rather than routing messages through other end devices.
4) **Limited Coverage:** Coverage depends on the radio range between the end devices and the coordinator.
5) **Low Complexity:** It requires relatively simple network management.
6) **Single Point of Failure:** If the coordinator fails, normal communication across the network is disrupted.

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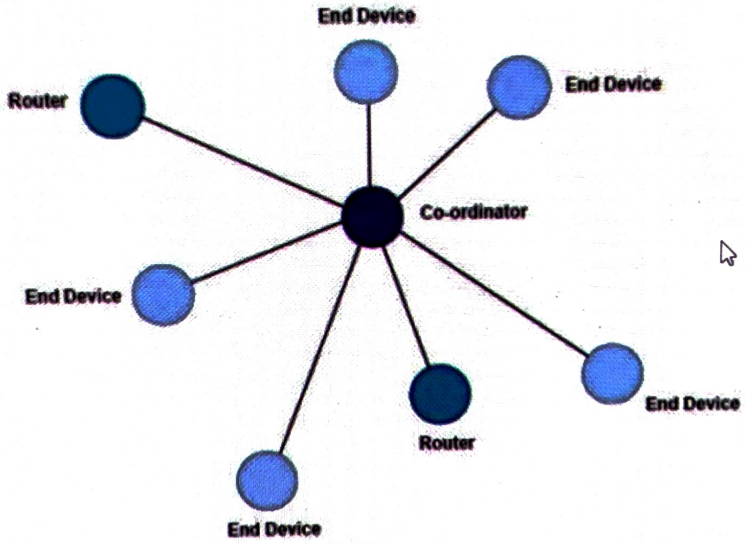
7) **Applications:** It is suitable for small smart-home systems, wireless switches, and simple sensor networks.

2. Tree Topology: In Tree topology, devices are arranged hierarchically. The coordinator acts as the root, routers form branches, and end devices connect as child devices.

Diagram:



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Characteristics:

1) **Hierarchical Structure:** Devices are arranged in parent-child relationships, with the coordinator at the root.
2) **Role of Routers:** Routers form branches and connect additional routers and end devices.
3) **Extended Coverage:** The network can cover a larger area than a single Star network by using routers.
4) **Organized Communication:** Data can travel through the hierarchy between devices and the coordinator.
5) **Scalability:** Additional branches and devices can be added within the limits of the network configuration.
6) **Dependence on Parent Devices:** Devices depend on their parent routers for connectivity.
7) **Failure Impact:** If a parent router fails, its child devices may lose connectivity until a route or parent connection is restored.
8) **Applications:** It is suitable for building automation, smart lighting systems, and multi-level sensor networks.

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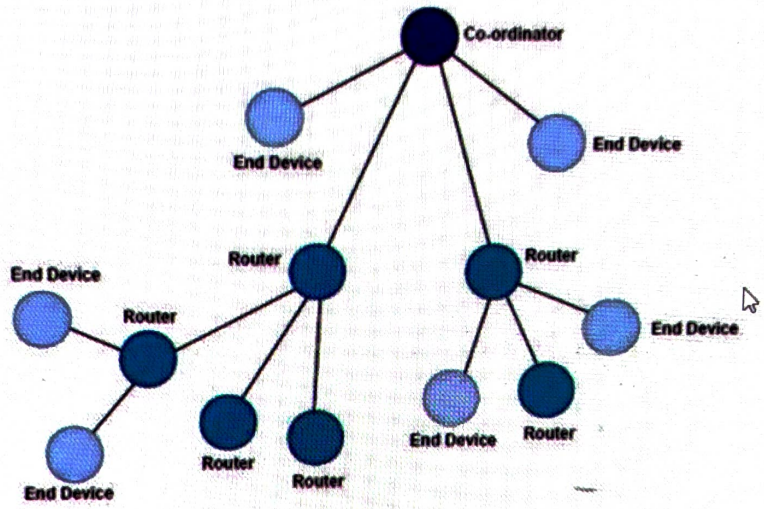
8) **Applications:** It is suitable for building automation, smart lighting systems, and multi-level sensor networks.

3. Mesh Topology: In Mesh topology, routers are interconnected so that messages can travel through multiple possible paths. This provides alternative routes when a link becomes unavailable.

Diagram:



Characteristics:

1) **Multiple Paths:** Data can travel through different routes to reach its destination.

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2) **Multi-Hop Communication:** Routers forward messages through intermediate devices when direct communication is not possible.

3) **High Reliability:** Alternative routes can help maintain communication if a link or router becomes unavailable.

4) **Self-Healing:** The network may find another available route when a communication path fails, depending on routing support and configuration.

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5) **Extended Coverage:** Interconnected routers can provide coverage over a wider area.
6) **Scalability:** Additional routers can be added to expand the network, subject to capacity and design constraints.
7) **Higher Complexity:** Routing and network management are more complex than in a simple Star topology.
8) **Power and Resource Requirements:** Routers must remain available to forward messages, so they generally require more resources than sleeping end devices.
9) **Applications:** It is suitable for industrial monitoring, large smart buildings, and extensive wireless sensor networks.

Thus, Zigbee uses Coordinator, Router, and End Device roles to manage network operations, while Star, Tree, and Mesh topologies provide different arrangements for connecting devices according to the required coverage, complexity, scalability, and reliability.

Q.10) Elaborate the need of new network architecture in IoT.

Ans: The **Internet of Things (IoT)** connects a large number of physical devices, sensors, machines, and applications through the Internet. Traditional network architectures were mainly designed for communication between computers and servers. However, IoT requires a more flexible and scalable network architecture to handle massive numbers of connected devices, continuous data generation, low-power communication, and real-time processing.

Need for New Network Architecture in IoT

1) **Massive Number of Connected Devices:** IoT networks connect billions of devices, including sensors, smart appliances, vehicles, and industrial machines. A new network architecture is required to support a large number of devices efficiently without overloading the network.
2) **Heterogeneous Devices and Networks:** IoT devices differ in hardware, processing capacity, operating systems, and communication technologies. The architecture must support different devices and networks, such as Wi-Fi, Bluetooth, Zigbee, cellular networks, and LPWAN.
3) **Large Volume of Data:** IoT devices continuously generate large amounts of data in different formats. The network architecture must support efficient data collection, transmission, storage, and processing.
4) **Real-Time Communication:** Applications such as healthcare monitoring, industrial automation, and traffic control require timely data transmission and quick responses. A suitable architecture helps reduce communication delays and supports real-time operations.
5) **Low Power Consumption:** Many IoT devices operate on batteries and are installed in remote locations. The network architecture must support energy-efficient communication to increase device battery life and reduce maintenance.
6) **Scalability and Flexibility:** The number of IoT devices may increase as an application grows. A new architecture should allow devices, gateways, services, and network resources to be added without requiring major changes to the entire system.
7) **Security and Privacy:** IoT devices exchange sensitive information and may control physical systems. The architecture must provide authentication, authorization, encryption, and secure device management to protect data and prevent

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5) **E. .ended Coverage:** Interconnected routers can provide coverage over a wider area.

6) **Scalability:** Additional routers can be added to expand the network, subject to capacity and design constraints.

7) **Higher Complexity:** Routing and network management are more complex than in a simple Star topology.

8) **Power and Resource Requirements:** Routers must remain available to forward messages, so they generally require more resources than sleeping end devices.

9) **Applications:** It is suitable for industrial monitoring, large smart buildings, and extensive wireless sensor networks.

Thus, Zigbee uses Coordinator, Router, and End Device roles to manage network operations, while Star, Tree, and Mesh topologies provide different arrangements for connecting devices according to the required coverage, complexity, scalability, and reliability.

Q.10) Elaborate the need of new network architecture in IoT.

Ans: The **Internet of Things (IoT)** connects a large number of physical devices, sensors, machines, and applications through the Internet. Traditional network architectures were mainly designed for communication between computers and servers. However, IoT requires a more flexible and scalable network architecture to handle massive numbers of connected devices, continuous data generation, low-power communication, and real-time processing.

Need for New Network Architecture in IoT

1) **Massive Number of Connected Devices:** IoT networks connect billions of devices, including sensors, smart appliances, vehicles, and industrial machines. A new network architecture is required to support a large number of devices efficiently without overloading the network.
2) **Heterogeneous Devices and Networks:** IoT devices differ in hardware, processing capacity, operating systems, and communication technologies. The architecture must support different devices and networks, such as Wi-Fi, Bluetooth, Zigbee, cellular networks, and LPWAN.
3) **Large Volume of Data:** IoT devices continuously generate large amounts of data in different formats. The network architecture must support efficient data collection, transmission, storage, and processing.
4) **Real-Time Communication:** Applications such as healthcare monitoring, industrial automation, and traffic control require timely data transmission and quick responses. A suitable architecture helps reduce communication delays and supports real-time operations.
5) **Low Power Consumption:** Many IoT devices operate on batteries and are installed in remote locations. The network architecture must support energy-efficient communication to increase device battery life and reduce maintenance.
6) **Scalability and Flexibility:** The number of IoT devices may increase as an application grows. A new architecture should allow devices, gateways, services, and network resources to be added without requiring major changes to the entire system.
7) **Security and Privacy:** IoT devices exchange sensitive information and may control physical systems. The architecture must provide authentication, authorization, encryption, and secure device management to protect data and prevent

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7) **Security and Privacy:** IoT devices exchange sensitive information and may control physical systems. The architecture must provide authentication, authorization, encryption, and secure device management to protect data and prevent unauthorized access.
8) **Interoperability:** Devices from different manufacturers may use different protocols and data formats. A new network architecture should support interoperability so that devices and applications can exchange information effectively.
9) **Reliable Communication:** IoT applications may operate in environments with unstable connectivity or network failures. The architecture should support reliable communication, fault handling, and recovery mechanisms wherever required.
10) **Support for Mobility:** Applications such as connected vehicles, asset tracking, and wearable devices involve devices that move between locations or networks. The architecture must support mobility and maintain connectivity when devices change their network attachment.
11) **Edge and Cloud Computing Integration:** Sending all IoT data to a centralized cloud can increase network traffic and delay responses. A modern architecture supports edge and fog computing to process data closer to devices while using cloud computing for large-scale storage and analytics.
12) **Efficient Network Management:** Managing a large number of devices manually is difficult. The architecture should support remote configuration, monitoring, software updates, fault detection, and device lifecycle management.
13) **Quality of Service (QoS):** Different IoT applications have different requirements for bandwidth, latency, reliability, and availability. The architecture should support suitable quality-of-service mechanisms according to application needs.
14) **Cost-Effective Deployment:** IoT systems may contain thousands of sensors distributed over large areas. A suitable network architecture should reduce communication, infrastructure, energy, and maintenance costs wherever possible.

Thus, a new network architecture is required in IoT to support massive connectivity, heterogeneous devices, efficient data handling, low power consumption, security, reliability, and real-time services. It enables IoT systems to remain scalable, flexible, and efficient as the number of connected devices increases.

Q.11) Compare and contrast Edge, Fog and Cloud Computing. Explain their usage with an example.

Ans: Edge, Fog, and Cloud Computing are computing models used in the Internet of Things (IoT) to process, store, and analyze data generated by connected devices. They differ mainly in the location of data processing, response time, computing resources, and communication requirements.

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1. **Edge Computing**: Edge computing processes data directly on or very close to the IoT device where the data is generated.

1) **Processing Location:** Data is processed on sensors, smart devices, cameras, or nearby controllers.
2) **Low Latency:** It provides quick responses because data does not need to travel to a distant server.
3) **Reduced Bandwidth:** Only necessary data or results are sent to the cloud, reducing network traffic.
4) **Local Decision-Making:** Devices can take immediate action based on locally processed data.
5) **Limited Resources:** Edge devices generally have less processing power and storage than fog or cloud systems.
6) **Example:** A smart security camera detects motion locally and immediately activates an alarm.

2. **Fog Computing**: Fog computing provides an intermediate computing layer between edge devices and the cloud. It processes and manages data through nearby gateways, routers, or local servers.

1) **Processing Location:** Data is processed on gateways, local servers, or other intermediate network devices.
2) **Data Aggregation:** It collects data from multiple edge devices and combines it before sending selected information to the cloud.
3) **Moderate Latency:** It generally responds faster than distant cloud processing, although slower than processing directly at the edge in many implementations.
4) **Regional Decision-Making:** It can coordinate several devices within a building, factory, campus, or local area.
5) **Resource Availability:** Fog nodes usually have more computing and storage capacity than simple edge devices but less than large cloud data centers.
6) **Example:** A factory gateway analyzes data from several machines and sends an alert when a machine shows abnormal vibration.

3) **Cloud Computing**: Cloud computing uses centralized remote data centers to provide computing power, storage, databases, and analytics services over a network.

1) **Processing Location:** Data is processed in remote data centers accessed through the Internet or another network.
2) **High Computing Capacity:** It provides powerful processors, large storage capacity, and advanced analytics services.
3) **Large-Scale Storage:** It stores data collected over long periods for historical analysis and reporting.
4) **Advanced Analytics:** It supports machine learning, artificial intelligence, forecasting, and analysis of data from many locations.
5) **Network Dependency:** Accessing cloud services requires network connectivity, and remote processing may introduce

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5) **Network Dependency:** Accessing cloud services requires network connectivity, and remote processing may introduce additional latency.

6) **Example:** A cloud platform analyzes months of factory sensor data to predict equipment failures and plan maintenance.

Comparison Between Edge, Fog and Cloud Computing

Sr. No.	Basis	Edge Computing	Fog Computing	Cloud Computing
1.	**Processing location**	On or very close to the IoT device.	On intermediate gateways or local servers.	In centralized remote data centers.
2.	**Distance from device**	Closest to the data source.	Located between edge devices and the cloud.	Usually farthest from the data source.
3.	**Latency**	Usually provides the lowest response time.	Provides relatively low latency for local or regional processing.	May have higher latency because data travels to remote servers.
4.	**Computing resources**	Usually limited by device hardware.	Moderate resources available at gateways or local servers.	Offers extensive and scalable computing resources.
5.	**Storage capacity**	Usually limited.	Moderate and suitable for local data aggregation.	Large and suitable for long-term storage.
6.	**Bandwidth usage**	Reduces traffic by processing data locally.	Reduces traffic by filtering and aggregating data before cloud transmission.	May require substantial data transmission from devices to the cloud.
7.	**Main purpose**	Immediate local processing and control.	Local coordination and processing across multiple devices.	Large-scale storage, analytics, and centralized management.

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8.	**Network dependency**	Can continue some operations without an Internet connection.	Can perform local functions during cloud disconnection if configured to do so.	Remote cloud services generally require network connectivity.
9.	**Scalability**	Depends on the number and capacity of edge devices.	Can be expanded by adding fog nodes and gateways.	Resources can be scaled across large data centers, subject to service limits and cost.
10.	**Typical applications**	Smart cameras, wearable devices, and immediate machine control.	Smart buildings, factory gateways, and local traffic management.	Predictive analytics, historical reporting, and large-scale IoT data analysis.

Example: Smart Traffic Management System

A smart traffic management system demonstrates how all three computing models work together.

- **Edge Computing:** Cameras and traffic sensors detect vehicle movement and estimate traffic density. Local controllers adjust traffic signals immediately when congestion is detected.
- **Fog Computing:** A nearby traffic-control server collects information from multiple intersections, identifies local congestion patterns, and coordinates signal timings across the area.
- **Cloud Computing:** The cloud stores traffic data over long periods and analyzes it to identify peak hours, predict congestion, and help authorities plan road improvements.

Thus, Edge Computing enables immediate local decisions, Fog Computing coordinates processing across nearby devices, and Cloud Computing provides large-scale storage and advanced analytics. Together, they support efficient, responsive, and scalable IoT applications.

Q.12) Explain the IoT enabled Smart Home automation and Smart Parking system with respect to sensors, Actuators, Framework, Protocol, data analysis and security.

Ans: IoT-Enabled Smart Home Automation

Smart Home Automation uses IoT technology to monitor and control household appliances such as lights, fans, air conditioners, door locks, and security systems through mobile applications or automatic rules.

Sr. No.	Component	Explanation
1.	**Sensors**	Temperature, motion, light, smoke, gas, and door sensors collect information about room

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Sr. No.	Component	Explanation
1.	**Sensors**	Temperature, motion, light, smoke, gas, and door sensors collect information about room conditions, human movement, and security. Humidity sensors and smart energy meters monitor indoor conditions and electricity consumption.
2.	**Actuators**	Relays control lights and fans, smart locks secure doors, motors operate curtains, and HVAC controllers regulate room temperature. Alarms and buzzers provide warnings during emergencies.
3.	**Framework**	The system consists of sensors, a controller such as ESP32, a gateway, a processing platform, actuators, and a mobile application. Sensor data is processed locally or in the cloud, and commands are sent to appliances automatically.
4.	**Protocols**	Wi-Fi connects devices to the Internet, Zigbee supports low-power communication, and BLE enables short-range connectivity. MQTT transfers sensor data and control messages, while HTTP/HTTPS supports communication with applications and web services.
5.	**Data Analysis**	Sensor readings are analyzed to control lighting, regulate temperature, monitor energy consumption, and detect unusual conditions. Historical data helps identify usage patterns and improve energy efficiency.
6.	**Security**	Authentication, encryption, access control, strong passwords, and secure firmware updates protect devices and user data. Network protection and security alerts help prevent unauthorized access.
7.	**Working Example**	When a motion sensor detects a person in a dark room, the controller switches on the light through a relay. If a smoke sensor detects smoke, the system activates an alarm and sends a notification to the homeowner.

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IoT-Enabled Smart Parking System

A Smart Parking System uses IoT technology to detect vacant parking spaces, monitor vehicle occupancy, and provide real-time parking information to drivers through mobile applications and electronic display boards.

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Sr. No.	Component	Explanation
1.	**Sensors**	Ultrasonic and infrared sensors detect vehicles in parking spaces. Magnetic sensors identify changes caused by vehicles, cameras monitor occupancy using image processing, and RFID readers identify authorized vehicles.
2.	**Actuators and Output Devices**	Barrier gates control vehicle entry and exit, LED indicators show parking-space status, and electronic display boards show vacant-space availability. Alarms and automated access-control devices support parking management.
3.	**Framework**	The system consists of parking sensors, a microcontroller, a gateway, a server or cloud platform, a database, and a mobile application. Sensor readings are processed to update parking availability, while controllers operate indicators and gates.
4.	**Protocols**	Wi-Fi connects cameras and controllers, Zigbee supports low-power sensor communication, and LoRaWAN provides long-range connectivity for suitable deployments. MQTT transfers parking-status updates, HTTP/HTTPS connects applications to services, and RFID supports vehicle identification.
5.	**Data Analysis**	Sensor data is analyzed to identify occupied and vacant spaces and calculate parking availability. Historical data helps identify peak hours, estimate parking demand, improve space utilization, and analyze vehicle flow.
6.	**Security**	Device authentication, encrypted communication, and access control protect the parking system. Secure payment processing, software updates, network monitoring, and audit logs help protect vehicle information and prevent unauthorized operation.
7.	**Working Example**	When a vehicle occupies a parking space, an ultrasonic sensor detects it and sends the data to the server. The system updates the parking database and display board, allowing drivers to locate other vacant spaces through a mobile application.

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Thus, Smart Home Automation uses IoT to improve household comfort, energy efficiency, and security, while Smart Parking helps manage parking spaces and vehicle movement. Both systems combine sensors, actuators, communication protocols, data analysis, and security to provide intelligent automated services.

Q.13) Explain effective Data Visualization methods with suitable example.

Ans: Data visualization is the graphical representation of data using charts, graphs, maps, and dashboards. In IoT, it helps users understand large amounts of sensor data, identify patterns, detect abnormal conditions, and make better decisions.

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Q.13) Explain effective Data Visualization methods with suitable example.

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Effective Data Visualization Methods

1) **Bar Chart:** A bar chart compares values across different categories. In IoT, it can compare the energy consumption of different appliances in a smart home.
2) **Line Chart:** A line chart displays changes in data over time. It is useful for monitoring temperature, humidity, air quality, and machine performance.
3) **Pie Chart:** A pie chart represents the proportion of different categories in a dataset. It can show the percentage of time a parking facility remains occupied or vacant.
4) **Histogram:** A histogram shows the frequency distribution of numerical data. It can display how frequently different temperature ranges occur in an industrial environment.
5) **Scatter Plot:** A scatter plot shows the relationship between two numerical variables. It can help analyze the relationship between machine temperature and energy consumption.
6) **Heat Map:** A heat map uses colours to represent data values. In a smart parking system, it can highlight parking areas with high and low occupancy.
7) **Real-Time Dashboard:** A dashboard displays important measurements and system status in one place. It can show current temperature, humidity, energy consumption, and device status in a smart building.
8) **Geographical Map:** A geographical map displays data according to location. It is useful for tracking connected vehicles, monitoring pollution levels, and locating IoT devices across a city.
9) **Gauge Chart:** A gauge chart displays a value against a defined range or threshold. It can show the current pressure, battery level, or machine temperature compared with its acceptable operating range.
10) **Alerts and Status Indicators:** Colours, symbols, and notifications highlight important conditions. For example, a red indicator can warn operators when a sensor reading exceeds a predefined safety limit.

Principles of Effective Data Visualization

1) **Clarity:** Use simple charts with clear titles, labels, and units.
2) **Accuracy:** Represent data correctly without misleading scales or proportions.

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3) **Relevance:** Select a visualization method that suits the type of data and purpose.

4) **Timeliness:** Update visualizations frequently enough for the application, especially when monitoring real-time IoT data.

5) **Consistency:** Use consistent colours, units, and formats across dashboards.

6) **Readability:** Avoid unnecessary decoration and excessive information.

Suitable Example: IoT-Based Smart Agriculture

In a smart agriculture system, sensors collect soil moisture, temperature, humidity, and water-level data.

1) A **line chart** displays changes in soil moisture over time.

2) A **gauge chart** shows the current soil moisture level against the required range.

3) A **bar chart** compares water consumption across different fields.

4) A **dashboard** displays all sensor readings and irrigation system status.

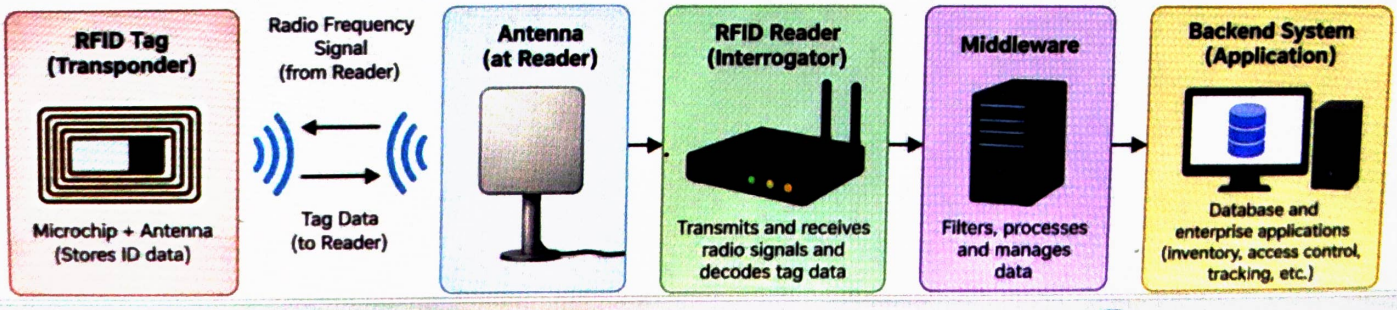
5) An **alert indicator** warns the farmer when soil moisture falls below the required threshold, allowing the irrigation system to be activated.

Thus, effective data visualization converts raw IoT data into understandable graphical information. Selecting suitable charts, maintaining accuracy, and presenting timely results helps users monitor systems, identify patterns, detect problems, and make informed decisions.

Q.14) Explain the architecture of RFID with its major applications.

Ans: Radio Frequency Identification (RFID) is a wireless technology used to identify and track objects using radio waves. It consists of RFID tags, a reader, antennas, and a host computer or application system. RFID is widely used in inventory management, access control, transportation, and supply chain management.

Architecture of RFID



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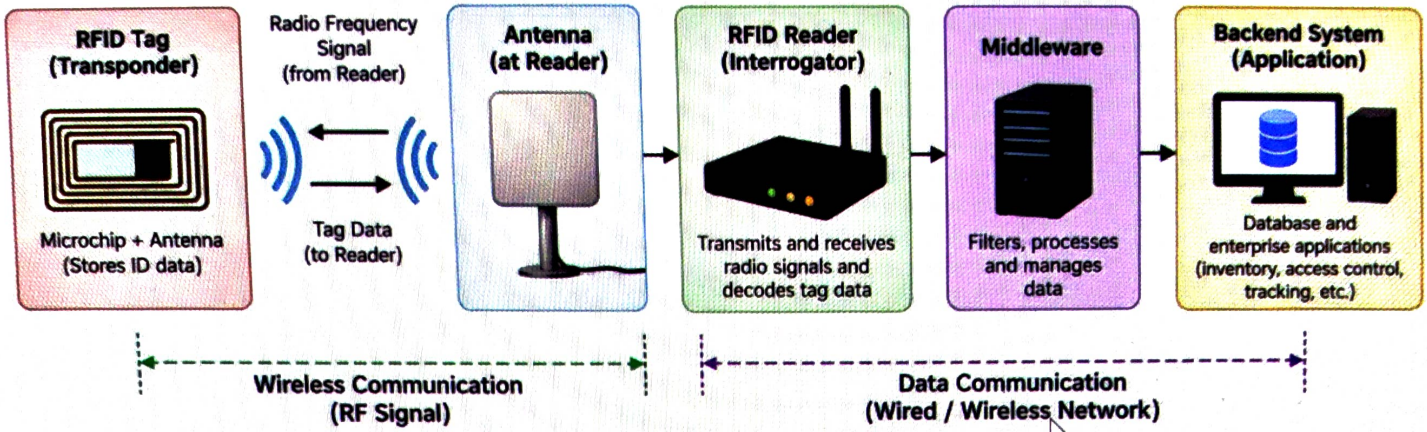
Architecture of RFID



1) **RFID Tag:** The tag is attached to an object and stores identification information. It consists of a microchip and an antenna. Tags may be passive, active, or battery-assisted, depending on their power source.
2) **RFID Reader:** The reader sends radio signals to communicate with tags and receives the identification data returned by them. It forwards the collected information to the processing system.
3) **Antenna:** The antenna transmits and receives radio-frequency signals between the reader and the tag. Its design and operating frequency affect the communication range.
4) **Host Computer or Application System:** It receives data from the reader, processes the tag identification information, and updates databases or business applications.
5) **Communication Network:** It connects the RFID reader to the host system through a suitable wired or wireless network, enabling data transfer and remote monitoring.

Working of RFID

1) The reader generates a radio-frequency signal through its antenna.
2) When an RFID tag enters the reader's operating range, it receives the signal. A passive tag uses energy from the reader's field to respond.
3) The tag transmits its stored identification information to the reader.
4) The reader receives and decodes the tag data.
5) The data is sent to the host computer or application system for processing.
6) The application identifies the object and performs the required action, such as updating inventory or verifying access.

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6) The application identifies the object and performs the required action, such as updating inventory or verifying access.

Major Applications of RFID

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1) **Inventory Management:** Identifies and tracks products in warehouses and retail stores.
2) **Access Control:** Uses RFID cards to grant authorized entry to buildings, offices, and laboratories.
3) **Supply Chain Management:** Tracks goods as they move between manufacturers, warehouses, and retailers.
4) **Transportation and Toll Collection:** Supports vehicle identification and automated toll collection using compatible RFID systems.
5) **Library Management:** Identifies books during issue, return, sorting, and stock verification.
6) **Healthcare:** Tracks medical equipment, supplies, and patient identification bands where suitable systems are deployed.
7) **Animal Identification:** Uses RFID tags to identify and track livestock and pets.
8) **Contactless Payments:** Supports payments and ticketing through compatible RFID or closely related contactless technologies.
9) **Asset Tracking:** Helps organizations locate and manage equipment, tools, and other valuable assets.
10) **Manufacturing:** Tracks components and products during production, assembly, and quality-control processes.

Thus, RFID enables automatic identification and tracking without requiring direct physical contact or line of sight in typical deployments. Its architecture allows tag data to be collected by a reader and processed by an application system for efficient monitoring and management.

Q.15) Draw and explain the architecture of Advanced Message Queuing Protocol (AMQP) with advantages.

Ans: The **Advanced Message Queuing Protocol (AMQP)** is an open-standard messaging protocol used for reliable communication between applications and devices. It enables messages to be transferred through message queues, allowing the sender and receiver to

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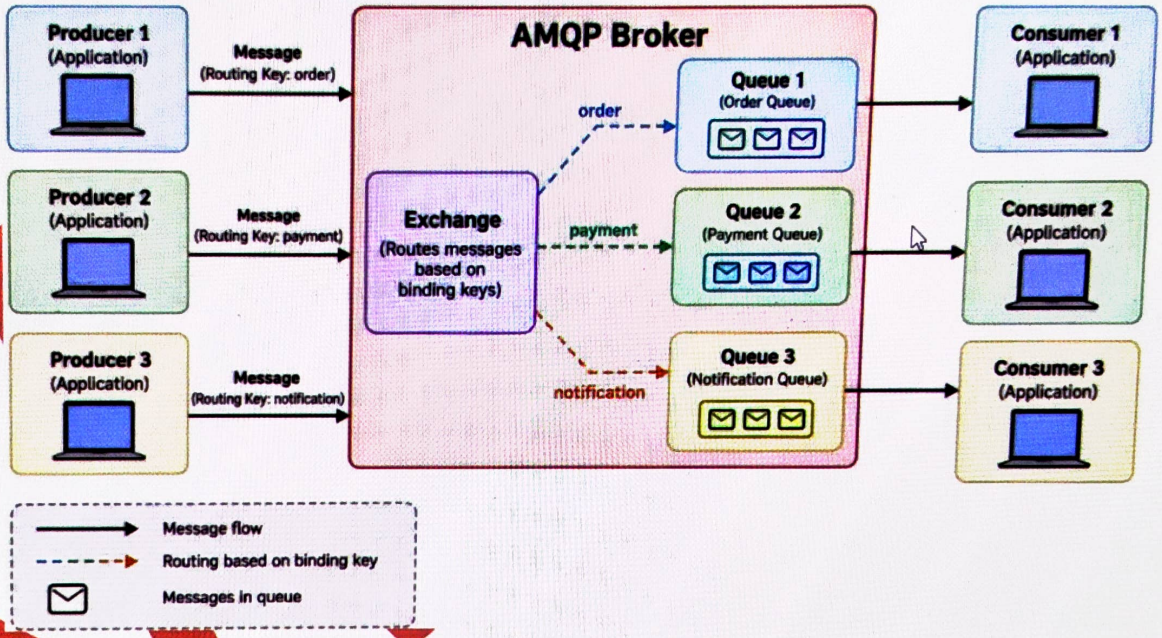
Q.15) Draw and explain the architecture of Advanced Message Queuing Protocol (AMQP) with advantages.

Ans: The **Advanced Message Queuing Protocol (AMQP)** is an open-standard messaging protocol used for reliable communication between applications and devices. It enables messages to be transferred through message queues, allowing the sender and receiver to communicate without being active at the same time. In IoT, AMQP is useful for secure, reliable, and asynchronous data exchange.

AMQP Architecture



The architecture of AMQP consists of the following components:

1) Producer: The producer is an application or device that generates messages and sends them to the AMQP broker. For example, an IoT sensor can send temperature readings to the broker.

2) Exchange: The exchange receives messages from producers and routes them to the appropriate queues based on routing rules. Different exchange types support different routing methods.

3) Binding: A binding connects an exchange to a queue and defines the routing conditions used to deliver messages to that queue.

4) Message Queue: A queue temporarily stores messages until they are delivered to consumers. It helps manage communication when producers and consumers operate at different times.

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communication when producers and consumers operate at different times.

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5) Consumer: The consumer receives messages from a queue and processes them. For example, a monitoring application may receive sensor readings and display them on a dashboard.

6) AMQP Broker: The broker manages exchanges, bindings, queues, and message delivery. It acts as an intermediary between producers and consumers, allowing them to communicate without being directly connected.

7) Acknowledgement: After processing a message, the consumer can send an acknowledgement to the broker. This helps the broker determine whether the message has been handled successfully.

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Working of the Architecture

1) The producer generates a message and sends it to the broker.
2) The exchange receives the message and checks the routing rules.
3) The message is directed to the appropriate queue through a matching binding.
4) The queue stores the message until a consumer receives it.
5) The consumer processes the message and may acknowledge successful processing.
6) The broker manages delivery according to the configured messaging and acknowledgement policies.

Advantages of AMQP

1) **Reliable Messaging:** Supports acknowledgements and delivery mechanisms that help prevent message loss.
2) **Asynchronous Communication:** Producers and consumers do not need to operate at the same time.
3) **Message Queuing:** Stores messages until they can be delivered to consumers, according to configuration.
4) **Flexible Routing:** Routes messages to appropriate queues based on routing keys and binding rules.
5) **Interoperability:** Supports communication between applications developed using different programming languages and

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5) **Interoperability:** Supports communication between applications developed using different programming languages and platforms.

6) **Security:** Supports authentication and can use TLS to protect communication.

7) **Scalability:** Multiple producers and consumers can communicate through a broker-based messaging system.

8) **Message Persistence:** Configured durable queues and persistent messages can help preserve messages across broker restarts.

9) **Load Distribution:** Messages can be distributed among multiple consumers to balance processing workloads.

10) **IoT Applications:** Suitable for industrial monitoring, telemetry processing, connected devices, and enterprise messaging systems.

Thus, AMQP provides a reliable and flexible messaging mechanism in which producers send messages through a broker to queues and consumers. Its routing, queuing, acknowledgement, and security features make it useful for IoT and distributed applications.

Q.16) Discuss the functional blocks of IoT architecture, highlighting their roles and interactions. Provide examples to illustrate the importance of each block in the overall functionality of IoT systems.

Ans: The functional blocks of IoT architecture are the major components that work together to collect data, communicate information, process data, control devices, and provide services to users. Each block performs a specific function, and their interaction enables an IoT system to operate efficiently, securely, and reliably.

Functional Blocks of IoT Architecture

1. Devices and Sensors: Devices interact with the physical environment by collecting information and performing actions.

1) Sensors measure physical parameters such as temperature, humidity, pressure, and motion.
2) Actuators perform physical actions according to control commands.
3) Microcontrollers process sensor readings and control connected components.
4) Devices may communicate directly with a network or through an IoT gateway.
5) These components form the connection between the physical world and the digital IoT system.

Example:

- In smart agriculture, a soil moisture sensor measures the moisture level in the soil.
- A water pump acts as an actuator and supplies water when the soil becomes too dry.

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2. Communication: The communication block enables data exchange between IoT devices, gateways, servers, and applications.

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1) It transfers sensor data to processing platforms and delivers control commands to devices.
2) It supports wired and wireless technologies according to the application's requirements.
3) Technologies such as Wi-Fi, Bluetooth, Zigbee, and cellular networks provide connectivity.
4) Protocols such as MQTT, HTTP, and CoAP support communication between devices and software services.
5) Gateways can connect devices that use different communication technologies.

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6) Reliable communication helps ensure that data reaches the correct destination.

Example:

- In a smart home, a temperature sensor sends room-temperature readings to an IoT controller through Wi-Fi or Zigbee.
- The controller sends a command to the air conditioner when cooling is required.

3. Services: The services block processes collected data and provides useful functions to IoT applications.

1) **Data Collection:** Receives readings from connected devices and sensors.
2) **Data Processing:** Converts raw readings into useful information.
3) **Event Handling:** Detects conditions that require a response.
4) **Device Discovery:** Helps identify available devices and their capabilities.
5) **Automation:** Executes predefined rules based on sensor readings or events.
6) **Data Storage and Retrieval:** Makes collected information available for later use when supported by the platform.

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Example:

- A smart home service checks the temperature received from a sensor.
- If the temperature exceeds the configured limit, it sends a command to switch on the air conditioner.

4. Management: The management block monitors, configures, and maintains IoT devices and system resources.

1) **Device Registration:** Registers new devices with the IoT platform.
2) **Configuration:** Sets device parameters and operating conditions.
3) **Monitoring:** Tracks device status, connectivity, and performance.
4) **Fault Management:** Identifies device failures and communication problems.
5) **Firmware Updates:** Supports remote software and firmware updates.
6) **Resource Management:** Helps manage network resources and connected devices as the system grows.

Example:

- An administrator monitors the status of sensors installed across a greenhouse.
- If a sensor stops responding, the management system identifies the problem and helps the administrator troubleshoot remotely.

5. Security: The security block protects IoT devices, communication, applications, and data against unauthorized access an attacks.

1) **Authentication:** Verifies the identity of users and devices.
2) **Authorization:** Determines which resources and operations a user or device can access.
3) **Encryption:** Protects data during transmission and storage.
4) **Data Integrity:** Helps detect unauthorized changes to information.
5) **Privacy Protection:** Prevents unauthorized exposure of sensitive user information.
6) **Secure Updates:** Helps ensure that device software and firmware are updated safely.
7) **Access Monitoring:** Records important security events and helps identify suspicious activity.

Example:

- A smart door lock verifies the identity of an authorized user before accepting an unlocking command.

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Example:

- A smart door lock verifies the identity of an authorized user before accepting an unlocking command.
- Encrypted communication helps protect commands exchanged between the mobile application and the smart lock.

6) Application: The application block provides user-oriented services for monitoring, analyzing, and controlling IoT systems.

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1) **Monitoring:** Displays current sensor readings and device status.
2) **Visualization:** Presents information through charts, graphs, and dashboards.
3) **Remote Control:** Allows users to control devices through mobile or web applications.
4) **Notifications:** Sends alerts when important conditions are detected.
5) **Reporting:** Presents historical data and system performance information.

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6) **Domain-Specific Services:** Supports applications in healthcare, agriculture, transportation, manufacturing, and smart homes.

Example:

- A smart agriculture application displays soil moisture, temperature, and humidity readings on a dashboard.
- A farmer can monitor irrigation activity and receive an alert when soil moisture falls below the required level.

Interaction Between Functional Blocks

The functional blocks cooperate to complete an IoT operation.

1) **Data Collection:** Devices and sensors collect information from the physical environment.
2) **Data Transmission:** The communication block transfers readings to the gateway or processing platform.

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Example:

- A smart door lock verifies the identity of an authorized user before accepting an unlocking command.
- Encrypted communication helps protect commands exchanged between the mobile application and the smart lock.

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3) **Remote Control:** Allows users to control devices through mobile or web applications.
4) **Notifications:** Sends alerts when important conditions are detected.
5) **Reporting:** Presents historical data and system performance information.

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6) **Domain-Specific Services:** Supports applications in healthcare, agriculture, transportation, manufacturing, and smart homes.

Example:

- A smart agriculture application displays soil moisture, temperature, and humidity readings on a dashboard.
- A farmer can monitor irrigation activity and receive an alert when soil moisture falls below the required level.

Interaction Between Functional Blocks

The functional blocks cooperate to complete an IoT operation.

1) **Data Collection:** Devices and sensors collect information from the physical environment.
2) **Data Transmission:** The communication block transfers readings to the gateway or processing platform.

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3) **Data Processing:** Services process the readings and evaluate predefined conditions.

4) **User Interaction:** The application displays the results and provides alerts to the user.

5) **Control Action:** A command is transmitted through the communication network to an actuator.

6) **System Maintenance:** The management block monitors device performance and helps resolve faults.

7) **Security Protection:** Security mechanisms protect devices, data, and communication throughout the operation.

Example: IoT-Based Smart Irrigation System

1) **Devices and Sensors:** A soil moisture sensor measures the water content in the soil.

2) **Communication:** The sensor sends its readings to an IoT controller or gateway.

3) **Services:** The platform compares the moisture reading with the predefined threshold.

4) **Application:** The mobile application displays soil conditions and irrigation status to the farmer.

5) **Control Action:** If the soil is too dry, the controller activates the water pump through a suitable switching mechanism.

6) **Management:** The system monitors the sensor and controller status to identify faults.

7) **Security:** Authentication and access control prevent unauthorized users from operating the irrigation system.

Thus, the functional blocks of IoT architecture work together to collect data, transfer and process information, provide user applications, manage devices, and maintain security. Their coordinated operation enables intelligent monitoring, automation, and control in IoT systems.

Q.17) Explain the importance of IoT data analytics.

Ans: IoT data analytics is the process of collecting, processing, and analyzing data generated by connected devices and sensors to obtain useful information. It helps identify patterns, detect abnormal conditions, predict future events, and make informed decisions. It converts raw IoT data into meaningful insights for improving efficiency and automation.

Importance of IoT Data Analytics

1) **Real-Time Monitoring:** IoT analytics processes incoming sensor data to monitor conditions continuously. It helps users observe changes in temperature, pressure, humidity, and machine performance.
   Example: A smart factory monitors machine temperature and alerts operators when it exceeds a safe limit.
2) **Better Decision-Making:** Analytics converts raw sensor readings into useful information for decision-making. It helps organizations make decisions based on actual data rather than assumptions.
   Example: Farmers use soil moisture readings to decide when irrigation is required.
3) **Predictive Maintenance:** IoT analytics examines machine performance data to identify early signs of equipment failure. It

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3) **Predictive Maintenance:** IoT analytics examines machine performance data to identify early signs of equipment failure. It helps schedule maintenance before a major breakdown occurs, reducing downtime and repair costs.
   Example: Vibration and temperature data can help predict possible failure in an industrial motor.
4) **Improved Operational Efficiency:** Analytics identifies inefficiencies in equipment, processes, and resource usage. It helps optimize operations, reduce wastage, and improve productivity.
   Example: A smart building analyzes electricity consumption to reduce unnecessary energy usage.
5) **Cost Reduction:** IoT analytics helps reduce maintenance, energy, and operational expenses. Early fault detection and efficient resource utilization can prevent unnecessary costs.
   Example: A connected irrigation system reduces water and electricity expenses by supplying water only when required.
6) **Automation and Intelligent Control:** Analytics enables systems to take actions automatically according to sensor readings and predefined rules. It reduces the need for continuous human intervention.
   Example: A smart home system automatically switches on an air conditioner when the room temperature exceeds the configured threshold.
7) **Improved Safety and Security:** Analytics detects unusual readings and suspicious activities in connected systems. It enables timely alerts and supports preventive action to reduce risks.

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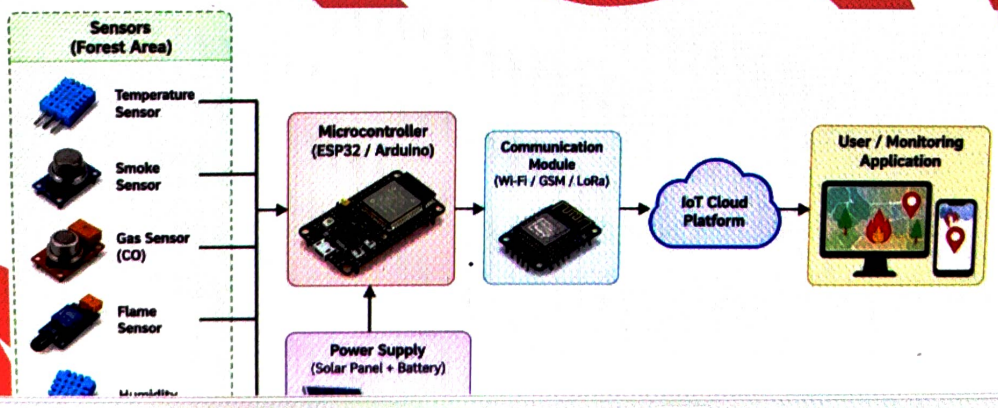
Example: Gas sensors in an industrial facility trigger an alert when gas concentration exceeds the permitted threshold.

8) **Better Customer Experience:** IoT analytics helps understand user behaviour, preferences, and usage patterns. It supports personalized services and improves the reliability of connected products.
   Example: A fitness wearable analyzes activity data and provides personalized activity summaries.
9) **Resource Optimization:** Analytics helps manage resources such as water, electricity, fuel, and raw materials efficiently. It reduces unnecessary consumption and supports sustainable operations.
   Example: Smart city systems analyze water consumption to identify unusual usage and possible leakage.
10) **Forecasting and Trend Analysis:** Historical IoT data helps identify trends and estimate future conditions. Forecasting supports planning, scheduling, and resource allocation.
   Example: A smart parking system analyzes historical occupancy data to estimate peak parking demand.
11) **Improved Product and Service Quality:** Analytics identifies recurring faults and performance problems in connected products. Manufacturers can use these insights to improve product design, reliability, and maintenance.
   Example: A manufacturer analyzes sensor data from connected machines to identify recurring performance issues.
12) **Remote Monitoring and Management:** IoT analytics enables organizations to monitor equipment and locations remotely. It reduces the need for frequent physical inspections and helps detect problems quickly.
   Example: An administrator monitors the condition of water pumps installed at remote locations through an IoT dashboard.

Thus, IoT data analytics is important because it converts raw sensor data into actionable information. It supports real-time monitoring, predictive maintenance, automation, cost reduction, improved safety, and better decision-making across smart homes, agriculture, healthcare, industries, and smart cities.

Q.18) Design the Forest Fire Detection system using IoT sensors.

Ans: An **IoT-based Forest Fire Detection System** uses sensors, microcontrollers, communication networks, and cloud platforms to detect early signs of forest fires. It continuously monitors environmental conditions and sends alerts to forest authorities when abnormal temperature, smoke, or gas levels indicate a possible fire.



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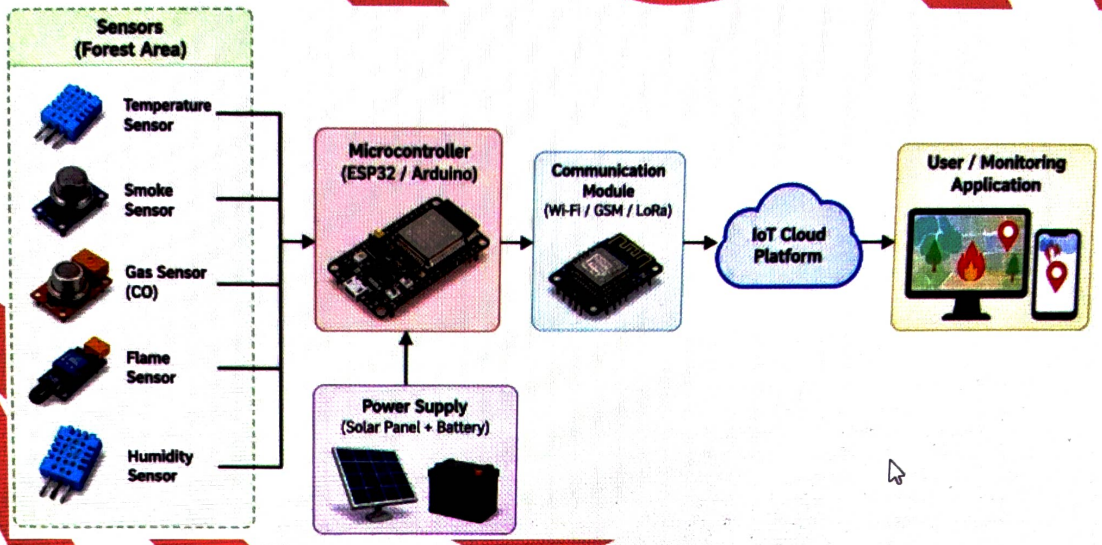

1. Sensors Used

1) **Temperature Sensor:** Measures the surrounding temperature and detects an unusual rise in heat.
2) **Smoke Sensor:** Detects smoke particles in the air that may indicate a fire.
3) **Gas Sensor:** Detects gases such as carbon monoxide produced during combustion.
4) **Humidity Sensor:** Measures atmospheric humidity because dry conditions can increase fire risk.
5) **Flame Sensor:** Detects infrared or ultraviolet radiation from flames within its sensing range.
6) **Wind Sensor:** Measures wind speed and direction to help assess the possible spread of fire.

2. Hardware Components

1) **Microcontroller:** ESP32 or Arduino processes sensor readings and controls the detection system.
2) **Communication Module:** Wi-Fi, cellular, LoRaWAN, or another suitable technology transmits alerts and sensor data.
3) **Power Supply:** Batteries, solar panels, or other suitable sources provide power to remote sensor nodes.
4) **Gateway:** Collects information from multiple sensor nodes and forwards it to the monitoring platform.
5) **Alarm System:** A buzzer or warning indicator can provide a local alert when configured conditions are detected.

3. Working of the System

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3. Working of the System

1) Sensors installed at different locations continuously measure temperature, smoke, gas concentration, and humidity.
2) The microcontroller collects readings and compares them with predefined thresholds.

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3) Multiple sensor readings can be checked together to reduce false alarms caused by isolated abnormal measurements.
4) When readings indicate a possible fire, the controller generates an alert and transmits the information through the available communication network.
5) The cloud platform or monitoring server records the sensor data and identifies the affected sensor location.
6) Forest authorities receive notifications through a dashboard, SMS, or mobile application, depending on the system configuration.
7) Authorities verify the alert and arrange inspection or emergency response as required.

4. Communication Technologies

1) **LoRaWAN:** Suitable for transmitting small amounts of sensor data over long distances with low power consumption when gateways are available.
2) **Cellular Networks:** Technologies such as 4G or LTE can send sensor readings and alerts where mobile coverage is available.
3) **Wi-Fi:** Can be used near forest stations or other locations with suitable Wi-Fi coverage.
4) **MQTT:** Transfers sensor readings and alerts to an IoT platform using lightweight messaging.

5. Data Processing and Analysis

1) Sensor readings are analyzed to identify abnormal environmental conditions.

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5. Data Processing and Analysis

1) Sensor readings are analyzed to identify abnormal environmental conditions.
2) Temperature, smoke, and gas readings can be combined to improve fire detection.
3) Historical data helps identify environmental patterns associated with increased fire risk.
4) GPS coordinates or registered sensor locations help identify the approximate location of a possible fire.
5) The system can prioritize alerts according to the severity and combination of detected conditions.

6. Security and Reliability

1) **Authentication:** Ensures that only registered devices can send data to the monitoring platform.
2) **Encryption:** Protects sensor readings and alerts during transmission.
3) **Fault Detection:** Identifies sensor failures, low battery levels, and communication problems.
4) **Backup Power:** Solar panels and batteries can support operation in remote locations.
5) **False Alarm Reduction:** Combining multiple sensor readings helps improve the reliability of detection.
6) **Communication Backup:** An alternative communication method can be provided where feasible to improve alert delivery.

Example: If a sensor node detects a sudden increase in temperature along with smoke and elevated carbon monoxide levels, the microcontroller evaluates the readings and sends a possible-fire alert to the monitoring platform. The platform identifies the sensor location and notifies forest authorities for verification and action.

Thus, an IoT-based Forest Fire Detection System supports continuous environmental monitoring and early warning. By combining sensors, communication technologies, data analysis, and alerts, it helps authorities respond more quickly to potential forest fires and reduce damage to forests and wildlife.

Q.19) Explain the functional blocks and the characteristics of IoT.

Ans: The **Internet of Things (IoT)** is a network of physical objects embedded with sensors, software, and communication capabilities that enable them to collect, exchange, and process data. IoT functional blocks define how the system operates, while its characteristics describe the important features that make IoT useful for monitoring, automation, and intelligent decision-making.

Functional Blocks of IoT

1) **Devices:** Devices include sensors, actuators, microcontrollers, and smart objects. Sensors collect physical data, such as temperature, pressure, and humidity. Actuators perform physical actions based on received commands.

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temperature, pressure, and humidity. Actuators perform physical actions based on received commands.

Example: A soil moisture sensor measures soil conditions, and a water pump supplies water when required.

2) **Communication:** This block enables data exchange between devices, gateways, servers, and applications. It uses technologies such as Wi-Fi, Bluetooth, Zigbee, and cellular networks. Protocols such as MQTT, HTTP, and CoAP support data transmission.

Example: A temperature sensor sends readings to a cloud platform through Wi-Fi.

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3) **Services:** Services process collected data and provide useful IoT functions. They support data collection, device discovery, event handling, and automation. They can trigger actions when predefined conditions are satisfied.

Example: A smart home service switches on an air conditioner when the room temperature exceeds a specified limit.

4) **Management:** Management handles device configuration, monitoring, and maintenance. It supports device registration, fault detection, and firmware updates. It helps maintain the performance and availability of IoT devices.

Example: An administrator remotely monitors sensors and updates their firmware.

5) **Security:** Security protects devices, communication, applications, and stored data. Authentication verifies the identity of users and devices. Encryption and access control help prevent unauthorized access and data exposure.

Example: A smart door lock verifies an authorized user before unlocking the door.

6) **Application:** This block provides user-oriented services for monitoring and controlling IoT systems. It displays sensor readings, device status, reports, and alerts. It supports applications in smart homes, healthcare, agriculture, industries, and smart cities.

Example: A smart agriculture application displays soil moisture readings and allows farmers to monitor irrigation remotely.

Characteristics of IoT

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Characteristics of IoT

1) **Connectivity:** IoT devices connect to networks to exchange data with other devices, gateways, and applications.
2) **Sensing:** Sensors collect information from the physical environment, such as temperature, humidity, pressure, and movement.
3) **Intelligence:** IoT systems process sensor data and use predefined rules or analytics to support intelligent decisions.
4) **Heterogeneity:** IoT includes different devices, hardware platforms, operating systems, and communication technologies that must work together.
5) **Dynamic Nature:** Device status, network conditions, location, and environmental readings can change continuously over time.
6) **Unique Identification:** Devices can be identified through IP addresses, device IDs, RFID tags, or other identification mechanisms, depending on the system.
7) **Scalability:** IoT architecture must support an increasing number of connected devices and growing volumes of data.
8) **Automation:** IoT systems can automatically perform actions based on sensor readings and predefined conditions, reducing human intervention.
9) **Self-Configuration:** Some IoT devices can automatically configure settings or join networks when supported by the system.
10) **Real-Time Data Processing:** IoT applications can process incoming data quickly to support timely monitoring, alerts, and control actions.
11) **Energy Efficiency:** Many IoT devices are designed to minimize power consumption, especially battery-powered sensors deployed in remote locations.
12) **Security and Privacy:** IoT systems require mechanisms to protect devices, communication, and sensitive data from unauthorized access.

Example of IoT Characteristics: Smart Home System

- **Connectivity:** Smart lights connect to a home network.
- **Sensing:** Temperature and motion sensors collect environmental data.
- **Intelligence:** The controller evaluates sensor readings and automation rules.
- **Automation:** Lights switch on when movement is detected in a dark room.
- **Security:** Authentication and encryption protect remote access and device communication.

Thus, the functional blocks of IoT define how devices collect data, communicate, process information, and provide services, while its characteristics enable connectivity, automation, scalability, and intelligent operation across different applications.

Q.20) Explain the role of HTTP, WebSocket, and MQTT in IoT communication. Compare and contrast these protocols in terms of their characteristics, suitability for different IoT scenarios, and support for real-time data transmission.

Ans: HTTP, WebSocket, and MQTT are application-layer communication protocols used in IoT systems to exchange data

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Q.20) Explain the role of HTTP, WebSocket, and MQTT in IoT communication. Compare and contrast these protocols in terms of their characteristics, suitability for different IoT scenarios, and support for real-time data transmission.

Ans: HTTP, WebSocket, and MQTT are application-layer communication protocols used in IoT systems to exchange data between devices, servers, and applications. They differ in their communication models, overhead, connection management, and suitability for real-time data transmission.

Role of HTTP, WebSocket, and MQTT in IoT

1. HTTP (Hypertext Transfer Protocol): HTTP is a request-response protocol in which a client sends a request and a server returns a response.

1) It is commonly used for communication between IoT devices, web servers, cloud platforms, and mobile applications.
2) It supports methods such as GET, POST, PUT, and DELETE to retrieve, send, update, and remove resources.

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3) It is suitable for device configuration, data uploads, retrieving sensor readings, and accessing REST APIs.
4) HTTP typically uses more communication overhead than MQTT for frequent, small sensor messages.
5) HTTPS provides encrypted communication using TLS.

Example: A smart electricity meter sends its energy readings to a cloud server through an HTTP POST request.

2. WebSocket: WebSocket is a protocol that establishes a persistent, full-duplex connection between a client and a server, allowing both sides to send data independently.

1) It supports continuous, two-way communication over a single connection.
2) It avoids the need to establish a separate HTTP request for every message after the WebSocket connection is established.
3) It is suitable for real-time dashboards, remote device control, live monitoring, and interactive applications.
4) It can provide low-latency communication when the network and server are responsive.

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5) WebSocket Secure (WSS) uses TLS to protect communication.

Example: A smart factory dashboard receives live machine-temperature updates through WebSocket while allowing an operator to send control commands.

3. MQTT (Message Queuing Telemetry Transport): MQTT is a lightweight publish-subscribe messaging protocol designed for efficient communication between IoT devices and applications. It commonly uses a broker to route messages between publishers and subscribers.

1) A **publisher** sends messages to a topic, and a **subscriber** receives messages from topics to which it subscribes.
2) The MQTT broker receives published messages and forwards them to matching subscribers.
3) It has low protocol overhead and is suitable for devices with limited bandwidth, processing power, or energy.
4) It supports three Quality of Service (QoS) levels: 0 (at most once), 1 (at least once), and 2 (exactly once at the MQTT protocol level).
5) Retained messages and persistent sessions can support particular delivery requirements when configured.
6) MQTT is widely used for sensor telemetry, remote monitoring, and IoT automation.

Example: Temperature sensors publish readings to a topic such as factory/temperature, and a monitoring application subscribes to that topic to receive updates.

Comparison of HTTP, WebSocket, and MQTT

Sr. No.	Basis	HTTP	WebSocket	MQTT
1)	**Communication model**	Uses a request-response model.	Supports full-duplex, two-way communication.	Uses a publish-subscribe model through a broker.
2)	**Connection**	Commonly uses separate request-response exchanges; connections may be reused.	Maintains a persistent connection after the handshake.	Commonly maintains a connection between clients and the broker.
3)	**Overhead**	Generally higher for frequent, small messages.	Low per-message overhead after connection establishment.	Very low protocol overhead, making it suitable for small messages.
4)	**Real-time support**	Suitable for periodic updates, but repeated requests may	Well suited to interactive, low-latency communication.	Well suited to continuous sensor updates and event-

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4)	**Real-time support**	Suitable for periodic updates, but repeated requests may introduce delay.	Well suited to interactive, low-latency communication.	Well suited to continuous sensor updates and event-driven messaging.
5)	**Communication direction**	Client initiates requests, and the server responds.	Both client and server can send messages independently.	Publishers send messages, and subscribers receive them through the broker.
6)	**Data delivery**	Application handles retries and other delivery requirements as needed.	Application must manage acknowledgements and delivery requirements when needed.	Provides QoS levels 0, 1, and 2 for different delivery guarantees.

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7)	**Intermediary**	Communicates with a web server or API endpoint.	Communicates with a WebSocket server.	Requires an MQTT broker for standard broker-based messaging.
8)	**Resource requirements**	Suitable for general-purpose web communication, but repeated requests may add overhead.	Requires resources to maintain persistent connections.	Designed for constrained devices and bandwidth-limited networks.
9)	**Best-suited applications**	Configuration, API access, data uploads, and reports.	Live dashboards, interactive control, and real-time browser updates.	Sensor telemetry, remote monitoring, and device-to-cloud messaging.
10)	**Security**	HTTPS uses TLS	WSS uses TLS	Supports TLS and client

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			updates.	cloud messaging.
10)	**Security**	HTTPS uses TLS.	WSS uses TLS.	Supports TLS and client authentication when configured.

Suitability for Different IoT Scenarios

- **HTTP:** Best suited for applications where devices upload data periodically, retrieve configurations, or communicate with cloud APIs. For example, a weather station uploads hourly measurements to a server.
- **WebSocket:** Best suited for applications requiring continuous, two-way communication and rapid interface updates. For example, an operator monitors industrial equipment and sends commands through a live dashboard.
- **MQTT:** Best suited for large numbers of IoT devices sending small messages over networks with limited bandwidth. For example, smart agriculture sensors publish soil-moisture readings to a broker, and an irrigation controller subscribes to the relevant topic.

Support for Real-Time Data Transmission

- **HTTP:** Can transmit data quickly, but traditional request-response communication is less convenient for continuous updates because the client generally initiates each exchange. Server-sent events or repeated requests can be used for updates, depending on the application.
- **WebSocket:** Supports low-latency, two-way communication over an established connection, making it suitable for live monitoring and interactive control.
- **MQTT:** Supports efficient, event-driven delivery of sensor data through a broker. Its latency depends on network conditions, broker performance, and QoS settings; it is not inherently a hard real-time protocol.

Thus, HTTP is suitable for request-response communication and web APIs, WebSocket is ideal for continuous two-way interaction, and MQTT is well suited to lightweight, event-driven IoT messaging. The appropriate protocol depends on device resources, network conditions, delivery requirements, and the need for real-time communication.

Q.21) Elaborate in detail the strategies to organize the data for IoT Analytics.

Ans: IoT devices generate large amounts of data in different formats and at high speeds. Data organization strategies help store, classify, process, and manage this data efficiently so that it can be analyzed to identify patterns and support decision-making.

1) **Data Collection and Aggregation:** Collect data from sensors, smart devices, machines, and gateways. Aggregate readings from multiple devices to reduce duplication and simplify analysis. Record essential information such as device ID, timestamp, and sensor values.
   Example: A smart city platform collects temperature and air-quality readings from sensors installed at different locations.
2) **Data Classification and Categorization:** Classify data according to its type, source, purpose, or application. Separate data

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Example: A smart city platform collects temperature and air-quality readings from sensors installed at different locations.

2) **Data Classification and Categorization:** Classify data according to its type, source, purpose, or application. Separate data into categories such as sensor readings, device status, location information, and system logs. Proper classification makes searching, filtering, and analysis easier.

Example: A smart factory stores machine temperature, vibration, and energy-consumption data in separate logical categories.

3) **Data Storage Management:** Select suitable storage systems according to data volume, format, access speed, and retention requirements. Relational databases store structured records, NoSQL databases support flexible data formats, and time-series databases are suitable for timestamped sensor readings. Cloud storage can support large-scale data storage.

Example: A smart energy meter stores timestamped electricity readings in a time-series database for analysis.

4) **Data Processing and Filtering:** Remove irrelevant or duplicate readings and validate incoming data to identify missing, invalid, or abnormal values. Process data at the edge or in the cloud according to application requirements. Filtering reduces unnecessary storage and communication overhead.

Example: A temperature-monitoring system filters duplicate readings before storing them for analysis.

5) **Data Integration:** Combine data from different sensors, devices, databases, and external sources. Standardize data formats, measurement units, and timestamps so that information from different sources can be analyzed together.

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Example: A smart agriculture system combines soil moisture, temperature, humidity, and weather forecast data to support irrigation decisions.

6) **Metadata Management:** Metadata describes the source, format, units, location, and collection time of data. Store device identifiers, sensor types, measurement units, and timestamps with relevant records. This helps users locate, understand, and interpret datasets correctly.

Example: A sensor reading of 35 becomes meaningful when its metadata identifies it as 35°C, recorded by a particular sensor at a specific location and time.

7) **Data Lakes and Data Warehouses:** A data lake stores large amounts of raw data in its original or near-original format, including structured, semi-structured, and unstructured data. A data warehouse stores organized and transformed data for reporting and structured analysis.

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reporting and structured analysis.

Example: A manufacturing company stores raw sensor logs in a data lake and summarized production statistics in a data warehouse.

8) **Data Retention and Archiving:** Define how long different categories of IoT data should be retained. Keep frequently accessed data available for current analysis and move older, less frequently accessed data to archival storage. Remove data according to applicable retention policies.
Example: A smart building keeps recent energy readings available for daily monitoring and archives older records for long-term analysis.

9) **Data Quality Management:** Check data for accuracy, completeness, consistency, and validity. Handle missing values, duplicate records, incorrect timestamps, and faulty sensor readings. Validation improves the reliability of analytics results.
Example: If a humidity sensor reports an impossible value, the system flags the reading for verification instead of treating it as reliable data.

10) **Real-Time and Batch Data Processing:** Real-time processing analyzes incoming data quickly to support immediate alerts and actions. Batch processing analyzes accumulated data at scheduled intervals to identify historical patterns and trends. The method depends on the application's response-time requirements.
Example: A forest-fire monitoring system processes sensor readings for immediate alerts and analyzes historical temperature and humidity data to assess fire risk.

11) **Data Visualization and Reporting:** Present organized data through charts, graphs, maps, and dashboards. Visualizations help users identify trends, compare values, detect abnormal conditions, and understand analytical results.
Example: A smart parking dashboard displays vacant spaces in real time and uses historical charts to identify peak parking hours.

12) **Data Security and Access Control:** Protect stored and transmitted data using encryption, authentication, and access-control mechanisms. Maintain backups and audit logs to support data recovery and accountability. Privacy measures help prevent unauthorized exposure of sensitive information.
Example: A healthcare IoT platform restricts patient-monitoring records to authorized personnel and protects data during transmission.

Thus, effective IoT data organization involves collecting, classifying, storing, filtering, integrating, and securing data. These strategies improve data quality and accessibility, enabling reliable analytics, timely decisions, and better IoT system performance.

Q.22) Explain the term in detail: i) Internet of Behaviour (IoB), ii) STOMP, iii) I-IoT

Ans:

i) **Internet of Behaviour (IoB):** The **Internet of Behaviour (IoB)** is a concept that combines IoT-generated data with behavioural analysis to understand people's activities, habits, interests, and preferences. It uses data collected from connected devices and digital systems to analyze behaviour and improve services, user experiences, and decision-making.

Characteristics and Features

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Characteristics and Features

1) **Data Collection:** Collects information from smart devices, wearable sensors, mobile applications, and online activities.
2) **Behavioural Analysis:** Analyzes collected data to identify habits, preferences, and behavioural patterns.
3) **Personalization:** Provides services and recommendations based on observed user behaviour.
4) **Predictive Analysis:** Uses historical data to estimate possible future actions or requirements.
5) **Decision Support:** Helps organizations improve services, marketing strategies, and operational decisions.
6) **Privacy and Security:** Requires appropriate consent, data protection, and access controls because behavioural data may be sensitive.

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Applications

1) **Healthcare:** Wearable devices analyze activity and sleep patterns to support health monitoring.
2) **Retail:** Shopping behaviour is analyzed to provide relevant product recommendations.
3) **Transportation:** Travel patterns help improve route planning and transportation services.

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4) **Smart Homes:** Device usage patterns help automate lighting, temperature, and energy consumption.
5) **Education:** Learning activity data can help identify learning preferences and improve educational services.

Example: A fitness wearable collects daily step counts and activity duration. An application analyzes these patterns and provides personalized activity suggestions to help the user achieve fitness goals.

ii) **STOMP (Simple Text Oriented Messaging Protocol): STOMP** is an application-layer messaging protocol that uses a simple text-based format for communication between clients and message brokers. It defines commands and frames that allow applications to send messages, subscribe to destinations, and receive messages through a messaging system.

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Characteristics and Features

1) **Text-Based Protocol:** Uses readable commands and headers, making messages relatively easy to understand and debug.
2) **Client-Broker Communication:** Enables clients to communicate with a compatible message broker.
3) **Messaging Commands:** Supports commands such as CONNECT, SEND, SUBSCRIBE, UNSUBSCRIBE, and DISCONNECT.
4) **Destination-Based Messaging:** Messages are sent to destinations, such as queues or topics, according to broker configuration.
5) **Acknowledgements:** Supports message acknowledgement mechanisms to indicate whether a message has been received or processed, depending on the configured mode.
6) **Interoperability:** Allows applications written in different programming languages to exchange messages through compatible brokers.
7) **Protocol Simplicity:** Its straightforward frame structure makes implementation and integration easier in suitable messaging applications.

Working of STOMP

1) A client establishes a connection with a STOMP-compatible broker.
2) The client sends a SEND frame to publish a message to a destination.
3) Another client sends a SUBSCRIBE frame to receive messages from that destination.
4) The broker routes messages to subscribed clients according to its configuration.
5) The receiving client processes the message and acknowledges it when required.

Applications

1) Real-time web messaging and notifications.
2) Communication between distributed applications.
3) Message exchange in enterprise systems.
4) Live dashboards and monitoring applications.
5) IoT applications that use a compatible messaging broker and require text-oriented messaging.

Example: In a smart building, a monitoring application can send temperature readings to a broker using STOMP. A dashboard

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Example: In a smart building, a monitoring application can send temperature readings to a broker using STOMP. A dashboard subscribed to the relevant destination receives the readings and displays the current temperature.

iii) I-IoT (Industrial Internet of Things): The **Industrial Internet of Things (IIoT)** refers to the use of connected sensors, machines, controllers, communication networks, and analytics in industrial environments. It enables industries to monitor equipment, automate processes, improve productivity, and make data-driven decisions.

Characteristics and Features

1) **Industrial Connectivity:** Connects machines, sensors, programmable logic controllers (PLCs), and industrial systems.
2) **Real-Time Monitoring:** Collects operational data such as temperature, pressure, vibration, and production rate.
3) **Predictive Maintenance:** Analyzes equipment data to identify early signs of failure and schedule maintenance.
4) **Automation and Control:** Supports automatic monitoring and control of industrial processes.
5) **Data Analytics:** Uses historical and real-time data to improve production quality and operational efficiency.
6) **Reliability:** Industrial systems are designed to support dependable operation under demanding environmental conditions.

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7) **Security:** Requires strong access control, network protection, and secure communication to protect industrial equipment and operations.
8) **Interoperability:** Integrates different machines, controllers, software platforms, and industrial communication technologies.

Applications

1) **Smart Manufacturing:** Monitors production lines and helps optimize manufacturing operations.

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2) **Predictive Maintenance:** Detects unusual vibration or temperature in machines before serious failures occur.
3) **Energy Management:** Monitors industrial energy consumption and identifies opportunities to reduce wastage.
4) **Supply Chain Monitoring:** Tracks materials, products, and equipment throughout industrial operations.
5) **Oil and Gas:** Monitors pipelines, pressure levels, and equipment conditions.
6) **Industrial Robotics:** Connects robots and controllers to coordinate production activities.

Example: In a manufacturing plant, vibration and temperature sensors monitor an electric motor. The IIoT platform analyzes the readings and detects unusual vibration patterns. It alerts maintenance staff so that the motor can be inspected before a major breakdown occurs.

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Thus, the Internet of Behaviour uses data to understand and analyze human behaviour, STOMP provides text-based messaging between clients and compatible brokers, and the Industrial Internet of Things connects industrial equipment for monitoring, automation, and improved operational efficiency.

Q.23) Explain the following access technologies with application area of each: i) IEEE 802.15.4 ii) LTE-A, 6LoWPAN

Ans:

i) **IEEE 802.15.4: IEEE 802.15.4** is a communication standard that defines the Physical (PHY) layer and Medium Access Control (MAC) layer for low-rate wireless personal area networks (LR-WPANs). It is designed for low-power, low-cost devices that transmit small amounts of data over short distances.

Characteristics

1) **Low Power Consumption:** Suitable for battery-operated sensors and devices.
2) **Low Data Rate:** Supports applications that transmit small amounts of data rather than high-bandwidth content.
3) **Low Cost:** Enables relatively inexpensive wireless sensor devices.
4) **Short-Range Communication:** Suitable for communication within homes, buildings, and local sensor networks.
5) **Multiple Frequency Bands:** Supports different frequency bands, including 2.4 GHz and region-dependent sub-GHz bands.
6) **Network Support:** Provides the foundation for technologies such as Zigbee and Thread, which add higher-layer networking functions.
7) **Channel Access:** Uses mechanisms such as CSMA/CA in applicable operating modes to help devices share the wireless channel.

Application Areas

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Application Areas

1) **Smart Homes:** Connects sensors, switches, and compatible smart appliances.
2) **Industrial Monitoring:** Collects data from low-power sensors measuring temperature, pressure, and vibration.
3) **Wireless Sensor Networks:** Connects multiple sensors to monitor environmental conditions.
4) **Smart Agriculture:** Supports soil-moisture and environmental sensors.
5) **Healthcare Monitoring:** Enables communication between suitable low-power monitoring devices.
6) **Building Automation:** Supports lighting control, occupancy sensing, and environmental monitoring.

Example: In a smart home, a Zigbee-compatible motion sensor based on IEEE 802.15.4 detects movement and sends information to a controller to activate the lights.

ii) **LTE-A (Long-Term Evolution Advanced): LTE-Advanced (LTE-A)** is an enhancement of LTE, a fourth-generation (4G) cellular mobile communication technology. It provides higher data rates, improved spectral efficiency, and greater network capacity, making it suitable for applications that require wide-area connectivity and reliable data transmission.

Characteristics

1) **High Data Rate:** Supports high-speed data transmission compared with earlier cellular technologies.

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2) **Carrier Aggregation:** Combines multiple component carriers to increase available bandwidth.
3) **Improved Spectral Efficiency:** Uses radio resources more efficiently to improve network capacity.
4) **Wide-Area Coverage:** Provides connectivity over large geographical areas through cellular infrastructure.
5) **Mobility Support:** Maintains connectivity for moving devices, including vehicles and mobile equipment.
6) **Low Latency:** Supports responsive communication, although actual latency depends on network conditions and

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6) **Low Latency:** Supports responsive communication, although actual latency depends on network conditions and configuration.

7) **Quality of Service (QoS):** Supports traffic prioritization and service requirements for different applications.

Application Areas

1) **Connected Vehicles:** Supports vehicle connectivity, fleet tracking, and communication with cloud services.

2) **Smart Cities:** Connects cameras, traffic-monitoring equipment, and other networked systems.

3) **Industrial IoT:** Provides cellular connectivity for remote monitoring and industrial equipment where appropriate.

4) **Video Surveillance:** Transmits video from remotely located cameras.

5) **Remote Monitoring:** Connects devices deployed across large geographical areas.

6) **Mobile Broadband:** Supports high-speed Internet access for mobile devices and connected equipment.

Example: A fleet management company uses LTE-A connectivity to transmit vehicle locations and operational information to a central monitoring platform.

iii) 6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks): 6LoWPAN is a networking technology that enables IPv6 packets to be transmitted over low-power wireless networks, particularly those based on IEEE 802.15.4. It uses adaptation mechanisms, including header compression and packet fragmentation, to accommodate IPv6 communication over networks with small frame sizes.

Characteristics

1) **IPv6 Support:** Enables low-power devices to communicate using IPv6 addressing.

2) **Header Compression:** Reduces the overhead of IPv6 and transport-layer headers to fit constrained wireless frames more efficiently.

3) **Packet Fragmentation:** Divides packets into smaller fragments when they exceed the available frame size.

4) **Low-Power Communication:** Supports battery-powered devices used in wireless sensor networks.

5) **Internet Connectivity:** Enables communication between constrained wireless devices and IPv6 networks through suitable network infrastructure.

6) **Mesh Support:** Can operate in network arrangements that support multi-hop communication, depending on the routing and networking solution.

7) **Interoperability:** Helps integrate resource-constrained wireless devices with IP-based systems.

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7) **Interoperability:** Helps integrate resource-constrained wireless devices with IP-based systems.

Application Areas

1) **Smart Homes:** Connects low-power sensors and control devices to IP-based networks.
2) **Smart Agriculture:** Supports communication between soil-moisture sensors and monitoring systems.
3) **Environmental Monitoring:** Connects sensors that measure temperature, humidity, and air quality.
4) **Smart Buildings:** Supports lighting, occupancy, and energy-monitoring devices.
5) **Industrial Sensor Networks:** Enables IP-based communication with suitable low-power monitoring devices.
6) **Smart Cities:** Connects distributed environmental and infrastructure sensors to IPv6 networks.

Example: In a smart agriculture system, a soil-moisture sensor communicates over an IEEE 802.15.4 wireless network using 6LoWPAN, allowing its IPv6 data to reach an IP-based monitoring application through a suitable gateway or network configuration.

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Thus, IEEE 802.15.4 provides low-power wireless communication at the PHY and MAC layers, LTE-A provides high-speed and wide-area cellular connectivity, and 6LoWPAN enables IPv6 communication over constrained wireless networks. The appropriate technology depends on the required range, data rate, power consumption, and network architecture.

Q.24) Differentiate between the following:

i) RFID with Bluetooth

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Sr. No.	Basis	RFID	Bluetooth
1.	**Full Form**	RFID stands for Radio Frequency Identification.	Bluetooth is a short-range wireless communication technology.

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Sr. No.	Basis	RFID	Bluetooth
1.	**Full Form**	RFID stands for Radio Frequency Identification.	Bluetooth is a short-range wireless communication technology.
2.	**Purpose**	RFID is mainly used to identify and track tagged objects.	Bluetooth is mainly used to exchange data between wireless devices.
3.	**Working Principle**	An RFID reader communicates with an RFID tag using radio waves.	Bluetooth-enabled devices establish a wireless connection to exchange data.
4.	**Frequency**	RFID operates at different frequency bands, such as LF, HF, and UHF.	Bluetooth operates in the 2.4 GHz ISM band.
5.	**Communication Range**	The range varies with the RFID frequency, tag type, reader, and environment.	Typical Bluetooth range is around 10 metres, but it varies with the device and Bluetooth class.
6.	**Power Requirement**	Passive RFID tags obtain energy from the reader's radio field and do not need an internal battery.	Bluetooth devices generally require a power source; BLE supports very low-power operation.
7.	**Data Transfer**	RFID generally transfers identification data and limited information from tags.	Bluetooth supports data exchange between connected devices, including audio and sensor data.
8.	**Device Pairing**	RFID systems generally do not require conventional device pairing.	Bluetooth devices commonly use connection procedures and may require pairing or authentication.
9.	**Applications**	It is used in inventory tracking, access cards, toll collection, and supply-chain management.	It is used in wireless headphones, keyboards, fitness trackers, and IoT sensors.
10.	**Typical Example**	An RFID reader scans a tagged product to identify it in a warehouse.	A smartphone connects to a Bluetooth-enabled fitness tracker to receive health and activity data.

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ii) Compare and contrast IoT data analytics and IoT network analytics.

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ii) Compare and contrast IoT data analytics and IoT network analytics.

Ans:

Sr. No.	Basis	IoT Data Analytics	IoT Network Analytics
1.	**Meaning**	It involves analysing data collected from IoT devices to extract useful information and insights.	It involves analysing network traffic and performance data to monitor and optimize IoT connectivity.
2.	**Main Objective**	Its main objective is to support decision-making, prediction, and automation.	Its main objective is to improve network performance, reliability, and security.
3.	**Data Analysed**	It analyses sensor readings, temperature, humidity, machine status, and user activity.	It analyses network traffic, bandwidth usage, latency, packet loss, and connection status.
4.	**Focus Area**	It focuses on understanding the physical environment, device behaviour, and operational patterns.	It focuses on communication between devices, gateways, servers, and cloud platforms.
5.	**Techniques Used**	It uses statistical analysis, machine learning, data mining, and predictive analytics.	It uses traffic analysis, anomaly detection, performance monitoring, and network diagnostics.
6.	**Tools Used**	It may use Python, Apache Spark, Hadoop, and cloud analytics platforms.	It may use network monitoring tools, packet analysers, SNMP-based monitoring, and network analytics platforms.
7.	**Output**	It produces trends, predictions, reports, and insights for decision-making.	It produces network performance reports, fault alerts, traffic patterns, and connectivity diagnostics.

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8.	**Applications**	It is used in predictive maintenance, smart agriculture, healthcare monitoring, and energy management.	It is used in network fault detection, congestion management, connectivity monitoring, and network security.
9.	**Example**	Analysing machine sensor data to predict equipment failure before it occurs.	Analysing network traffic to detect packet loss or congestion affecting IoT devices.
10.	**Relationship**	It helps improve business operations and decisions using the information generated by IoT devices.	It helps ensure that IoT devices can communicate reliably so that data can be transmitted and accessed effectively.

iii) Physical and logical design of IoT

Ans:

Sr. No.	Basis	Physical Design of IoT	Logical Design of IoT
1.	**Meaning**	Physical design describes the hardware components and communication technologies used in an IoT system.	Logical design describes the functional organization and working of an IoT system.
2.	**Components**	It includes sensors, actuators, controllers, devices, and communication interfaces.	It includes functional blocks, communication models, and communication APIs.
3.	**Focus**	It focuses on the physical structure and connectivity of IoT devices.	It focuses on how different components interact and exchange information.
4.	**Nature**	It mainly deals with hardware and communication infrastructure.	It mainly deals with software functions, data flow, and communication mechanisms.
5.	**Sensors and Actuators**	It specifies the sensors used to collect data and actuators used to perform physical actions.	It describes how sensor data is processed and how commands are sent to actuators.
6.	**Communication**	It identifies communication technologies such as Wi-Fi, Bluetooth, Zigbee, and RFID.	It defines communication models such as request-response and publish-subscribe.
7.	**APIs**	It involves physical communication interfaces and device connections.	It includes APIs such as REST-based APIs and WebSocket APIs for communication between

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7.	**APIs**	It involves physical communication interfaces and device connections.	It includes APIs such as REST-based APIs and WebSocket APIs for communication between applications and services.
8.	**Representation**	It can be represented using hardware block diagrams showing devices and their connections.	It can be represented using functional diagrams, data-flow diagrams, and communication models.
9.	**Example**	A temperature sensor, ESP32 controller, Wi-Fi module, and air conditioner form the physical components of a smart home.	The logic that reads temperature, sends data to an application, compares it with a threshold, and controls the air conditioner represents the logical design.
10.	**Purpose**	Its purpose is to provide the hardware and connectivity required for an IoT system.	Its purpose is to define how the IoT system operates and how its components work together.

Q.25) Write a short note on:

i) Explain data acquisition in IoT

Ans: Data acquisition in the Internet of Things (IoT) is the process of collecting physical-world data using sensors and devices, converting it into digital form, and transferring it to a processing system for monitoring, analysis, and decision-making.

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Steps Involved in Data Acquisition in IoT:

1) **Data Sensing:** Sensors measure physical parameters such as temperature, humidity, pressure, light, motion, and air quality.
2) **Signal Conditioning:** The collected signals are filtered, amplified, or adjusted to remove noise and make them suitable for further processing.
3) **Data Conversion:** An Analog-to-Digital Converter (ADC) converts analog sensor signals into digital values when required.

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4) **Data Processing:** A microcontroller or processor processes the collected data, performs basic calculations, and may remove invalid or unnecessary readings.

5) **Data Transmission:** The processed data is transmitted to a gateway, edge device, or cloud platform using technologies such as Wi-Fi, Bluetooth, Zigbee, or LoRaWAN.

6) **Data Storage:** The collected data is stored in a database or cloud storage system for future access and analysis.

7) **Data Analysis and Action:** The system analyses the data and generates alerts or control actions based on predefined conditions.

Example: Smart Agriculture

In a smart irrigation system, a soil-moisture sensor measures the moisture level in the soil. The microcontroller collects and processes the readings and sends them to an IoT platform. If the moisture level falls below a predefined threshold, the system can automatically switch on the water pump.

Importance of Data Acquisition in IoT:

1) It enables real-time monitoring of physical conditions.
2) It provides data for analytics and intelligent decision-making.
3) It supports automation and remote control.
4) It helps detect abnormal conditions and generate alerts.
5) It improves the efficiency of systems such as smart homes, healthcare, and industrial monitoring.

Thus, data acquisition is a fundamental process in IoT because it converts real-world measurements into useful digital information for monitoring, analysis, and automation.

ii) Data retention strategy

Ans: A data retention strategy in IoT determines how long collected data should be stored, where it should be stored, and when it should be archived or deleted. It helps manage large volumes of IoT data efficiently, reduce storage costs, and protect sensitive information.

Steps Followed in a Data Retention Strategy:

1) **Data Classification:** IoT data is classified according to its type, importance, sensitivity, and usage. For example, sensor readings, system logs, and personal data may require different retention periods.
2) **Defining Retention Period:** A specific period is decided for storing each type of data according to its usefulness and

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2) **Defining Retention Period:** A specific period is decided for storing each type of data according to its usefulness and requirements. Frequently used data may be retained for a shorter period, while important historical records may be kept longer.

3) **Selecting Storage:** Suitable storage systems, such as edge devices, cloud databases, or data lakes, are selected based on data volume, accessibility, and cost.

4) **Data Archiving:** Older data that is not frequently accessed but may be required for future analysis is transferred to long-term or low-cost storage.

5) **Data Deletion:** Data that is no longer required is securely deleted after its retention period, provided it is not needed for legal, operational, or analytical purposes.

6) **Ensuring Data Security:** Encryption, access controls, and secure deletion methods are used to prevent unauthorized access and protect sensitive information.

7) **Reviewing and Monitoring:** Retention policies are reviewed regularly to ensure that they meet organizational requirements, applicable laws, and security standards.

Example: Smart Agriculture

In a smart agriculture system, soil-moisture readings are stored for irrigation control. Daily summaries may be retained for several months to analyze crop conditions, while important seasonal records are archived for future comparison. Unnecessary data is deleted according to the retention policy.

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Advantages of a Data Retention Strategy:

1) It reduces unnecessary data storage costs.
2) It improves data organization and retrieval.
3) It supports historical analysis and decision-making.
4) It protects sensitive information and supports privacy.

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5) It helps meet legal and organizational requirements.

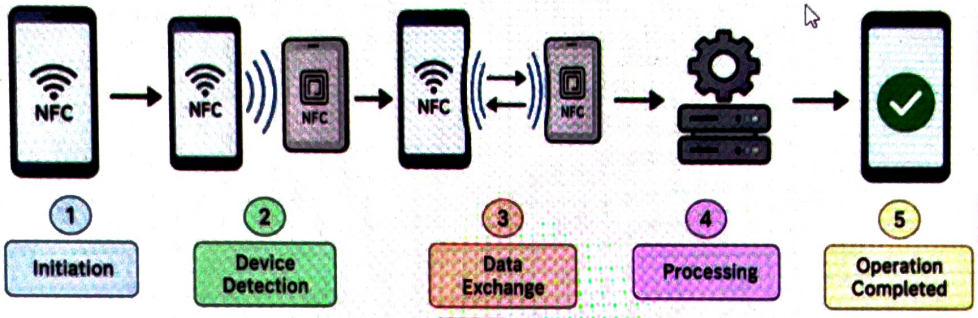
6) It prevents the indefinite storage of unnecessary data.

Thus, a data retention strategy ensures that IoT data is stored for the required period, archived when necessary, and securely deleted when it is no longer needed.

iii) Near Field Communication

Ans: Near Field Communication (NFC) is a short-range wireless communication technology that allows two compatible devices to exchange data when they are brought close to each other, usually within a distance of about 4 cm. It operates at a frequency of 13.56 MHz and is commonly used for contactless payments, access control, and data sharing.

Working of NFC:



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1) **Initiation:** An NFC-enabled device, such as a smartphone or card reader, initiates communication with another compatible device.

2) **Device Detection:** The devices are brought close together, generally within a few centimetres, to establish communication.

3) **Data Exchange:** The devices exchange small amounts of information using electromagnetic induction.

4) **Processing:** The receiving device processes the received information to perform an operation, such as completing a payment or verifying access.

Modes of Operation:

1) **Reader/Writer Mode:** An NFC device reads data from or writes data to an NFC tag. For example, a smartphone reads information from an NFC-enabled poster.

2) **Peer-to-Peer Mode:** Two NFC-enabled devices exchange information with each other. For example, compatible devices exchange contact information.

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3) **Card Emulation Mode:** An NFC device behaves like a contactless smart card. For example, a smartphone is used to make a contactless payment.

Applications of NFC:

1) **Contactless Payments:** NFC enables users to make payments using smartphones, smartwatches, or contactless cards.
2) **Access Control:** NFC cards or devices are used to unlock doors and enter offices, colleges, or buildings.
3) **Public Transportation:** NFC-enabled cards and devices are used for ticketing and fare payments.
4) **Smart Posters:** NFC tags provide information or open websites when a compatible smartphone is tapped against them.
5) **Device Pairing:** NFC helps initiate connections between compatible devices, such as smartphones and speakers.
6) **Identification:** NFC-enabled cards can store information used for identification and authentication.

Example: In a contactless payment system, a user brings an NFC-enabled smartphone close to a payment terminal. The devices exchange the required payment information, and the transaction is processed after the necessary authentication.

Thus, NFC is a short-range wireless technology that enables convenient data exchange and contactless transactions in IoT and everyday applications.

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iv) Bluetooth Low Energy (BLE) and its application

Ans: Bluetooth Low Energy (BLE) is a wireless communication technology designed to exchange small amounts of data over short distances while consuming very little power. It is widely used in IoT devices such as fitness trackers, smart sensors, medical devices, and smart home systems. BLE operates in the 2.4 GHz ISM frequency band.

Working of Bluetooth Low Energy (BLE):

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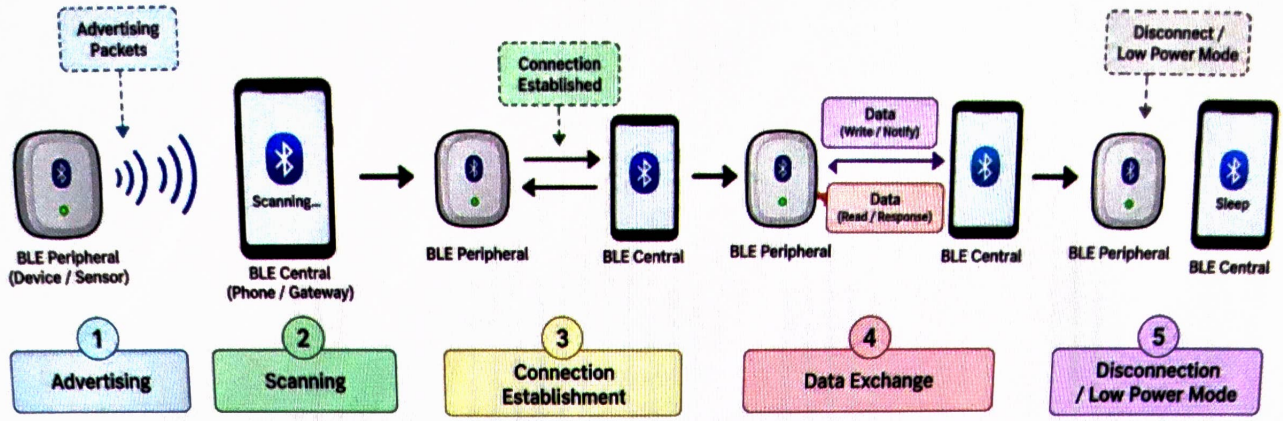

1) **Advertising:** A BLE device broadcasts small packets of information to announce its presence and services.
2) **Scanning:** Another BLE-enabled device scans for these advertising packets and detects the available device.
3) **Connection Establishment:** The devices establish a connection when required to exchange data securely and reliably.
4) **Data Exchange:** The connected devices exchange data using the Generic Attribute Profile (GATT), which organizes information into services and characteristics.
5) **Disconnection and Sleep:** After data exchange, devices may disconnect or enter a low-power state to conserve battery energy.

Applications of Bluetooth Low Energy (BLE):

1) **Fitness Trackers and Wearables:** BLE transfers step counts, heart-rate readings, and activity information from wearable devices to smartphones.
2) **Smart Home Automation:** BLE connects devices such as smart bulbs, locks, and sensors for monitoring and control.
3) **Healthcare Monitoring:** BLE-enabled medical devices transmit measurements such as blood glucose levels and body temperature to compatible devices.
4) **Asset Tracking:** BLE beacons help identify and locate objects such as equipment, luggage, and inventory within a particular area.
5) **Smart Retail:** BLE beacons can provide location-based offers and product information to nearby smartphones.
6) **Industrial IoT:** BLE sensors monitor parameters such as temperature, vibration, and equipment status in industrial environments.
7) **Proximity Detection:** BLE helps detect nearby devices and supports indoor location-based services.

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7) **Proximity Detection:** BLE helps detect nearby devices and supports indoor location-based services.

Advantages of BLE:

1) It consumes very little power and supports long battery life.
2) It is suitable for small, low-cost IoT devices.
3) It is supported by many smartphones, computers, and embedded devices.
4) It allows wireless communication without physical cables.
5) It supports communication with sensors and wearable devices.

Example: Fitness Tracker

A fitness tracker measures a user's physical activity and heart rate. It sends the collected readings to a smartphone using BLE, allowing the user to view activity data through a mobile application.

Thus, Bluetooth Low Energy is an energy-efficient wireless technology that enables battery-powered IoT devices to communicate and exchange data effectively.

Q.26) Consider Smark smoke, Smart farming system, Smart Irrigation system, Elaborate its working with block diagram and list down the different types of sensors and actuators required during the deployment scenario.

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Ans:

1. Smart Smoke Detection System: A Smart Smoke Detection System is an IoT-based safety system that detects smoke or fire-related conditions using sensors and sends alerts to users. It can activate alarms and other safety mechanisms when the measured values exceed predefined thresholds.

Working:

1) **Sensing:** Smoke and gas sensors continuously monitor the surrounding environment. Temperature sensors may also be used to detect abnormal increases in temperature.

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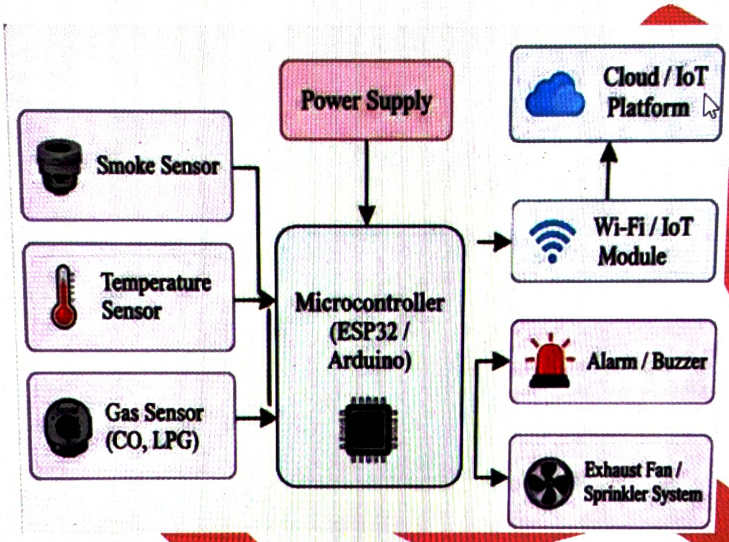
2) **Data Processing:** A microcontroller, such as an ESP32 or Arduino, collects sensor readings and compares them with predefined threshold values.

3) **Communication:** When smoke or abnormal conditions are detected, the controller sends an alert through Wi-Fi, GSM, or another suitable communication technology.

4) **Alert Generation:** A buzzer or alarm is activated, and a notification can be sent to the user's mobile phone or monitoring dashboard.

5) **Safety Action:** If the system is designed for it, suitable actuators can activate ventilation or other safety mechanisms.

Block Diagram:



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Sensors Required:

1) **Smoke Sensor:** Detects smoke particles in the surrounding air.

2) **Gas Sensor:** Detects certain combustible or hazardous gases, depending on the sensor type.

3) **Temperature Sensor:** Measures temperature and helps identify unusual heat conditions.

Actuators Required:

1) **Buzzer or Alarm:** Produces a sound when smoke or a dangerous condition is detected.

2) **Exhaust Fan:** Helps remove smoke from an area when ventilation is appropriate and safe.

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2) **Exhaust Fan:** Helps remove smoke from an area when ventilation is appropriate and safe.

3) **Relay Module:** Switches compatible external devices, such as a fan or alarm, under controller command.

Example: In a college laboratory, the system detects smoke, activates an alarm, and sends a notification to the responsible staff.

2. Smart Farming System: A Smart Farming System uses IoT sensors, communication technologies, and data analysis to monitor crop and field conditions. It helps farmers make informed decisions about crop health, soil conditions, weather, and resource usage.

Working:

1) **Data Collection:** Sensors collect information about soil moisture, temperature, humidity, light intensity, and other field conditions.

2) **Data Processing:** A microcontroller or IoT gateway gathers the readings and processes them.

3) **Data Transmission:** The collected data is transmitted through Wi-Fi, LoRaWAN, or cellular communication to an IoT platform.

4) **Data Analysis:** The platform displays field conditions and helps identify problems such as unsuitable temperature, low soil moisture, or poor environmental conditions.

5) **Control and Monitoring:** Farmers can monitor the field remotely and operate compatible devices, either manually or automatically, based on sensor readings.

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Block Diagram:

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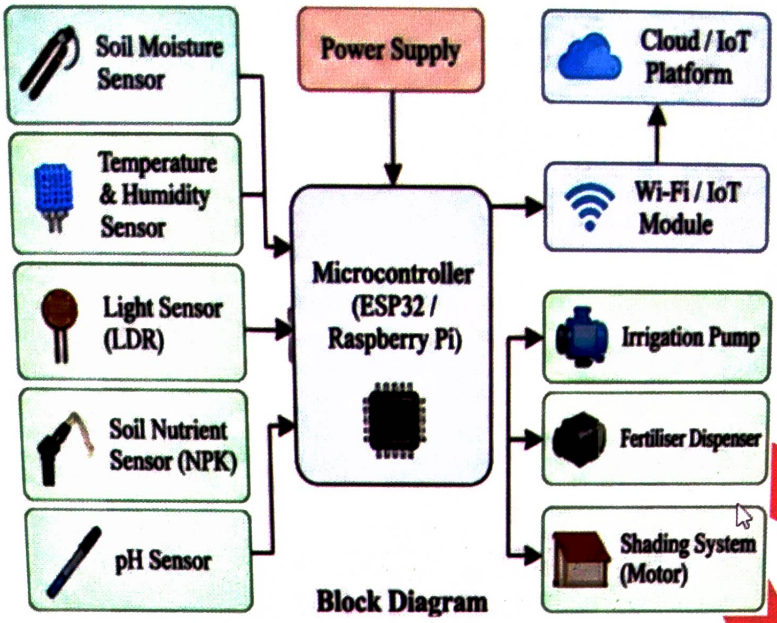

Block Diagram

Sensors Required:

1) **Soil Moisture Sensor:** Measures the amount of moisture present in the soil.
2) **Temperature Sensor:** Measures air or soil temperature.
3) **Humidity Sensor:** Measures the moisture level in the air.
4) **Light Sensor:** Measures sunlight intensity available to crops.
5) **Soil pH Sensor:** Measures soil acidity or alkalinity.
6) **Soil Nutrient Sensor:** Estimates selected soil nutrient levels, depending on the sensor technology.
7) **Rain Sensor:** Detects rainfall or the presence of water.

Actuators Required:

1) **Water Pump:** Supplies water to the field when required.
2) **Solenoid Valve:** Controls water flow through irrigation pipes.
3) **Motorized Sprayer:** Operates a spraying mechanism where automated spraying is required.
4) **Grow Lights:** Provide supplemental light in controlled farming environments, such as greenhouses.
5) **Ventilation Fan:** Regulates airflow in a greenhouse.

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Actuators Required:

1) **Water Pump:** Supplies water to the field when required.
2) **Solenoid Valve:** Controls water flow through irrigation pipes.
3) **Motorized Sprayer:** Operates a spraying mechanism where automated spraying is required.
4) **Grow Lights:** Provide supplemental light in controlled farming environments, such as greenhouses.
5) **Ventilation Fan:** Regulates airflow in a greenhouse.

Example: A farmer monitors soil moisture, temperature, and humidity through a mobile dashboard and uses the readings to decide when to irrigate or take other crop-management actions.

3. Smart Irrigation System: A Smart Irrigation System is an IoT-based system that automatically or remotely controls the supply of water to crops according to soil moisture and other environmental conditions. Its main objective is to provide adequate water while reducing wastage.

Working:

1) **Soil Monitoring:** A soil moisture sensor measures the moisture content of the soil near the plant roots.
2) **Data Collection:** The sensor sends its readings to a microcontroller.
3) **Decision Making:** The controller compares the measured moisture level with a predefined threshold. It may also consider temperature or rainfall data.
4) **Actuator Activation:** If the soil is too dry and watering is appropriate, the controller switches on the water pump or opens a solenoid valve.
5) **Automatic Stopping:** When the required moisture level is reached, the controller switches off the pump or closes the valve.
6) **Remote Monitoring:** Sensor readings and irrigation status can be displayed on an IoT dashboard or mobile application.

Block Diagram:

Sejal
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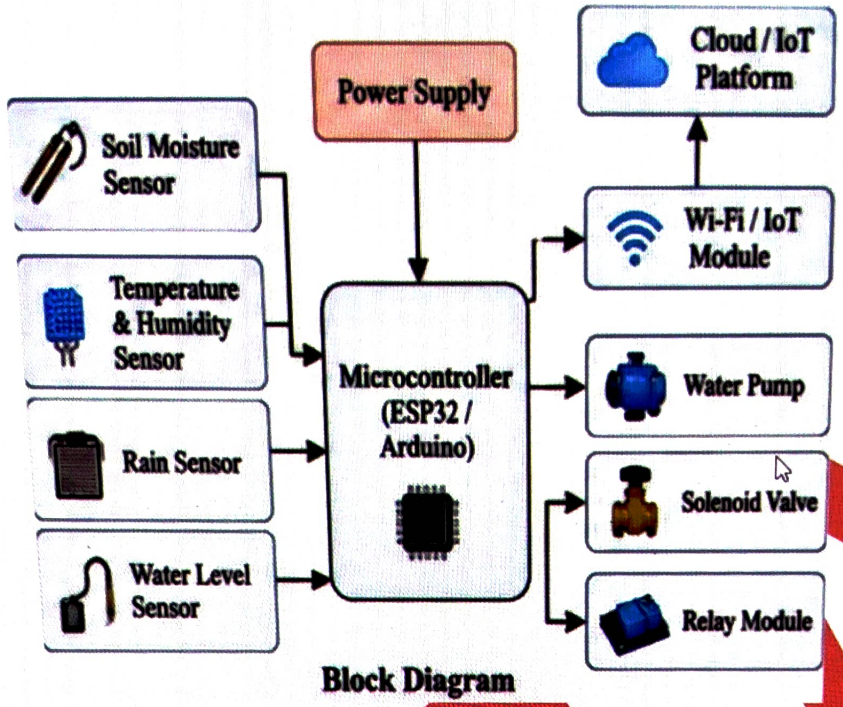

Block Diagram

Gaikwad
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Sensors Required:

1) **Soil Moisture Sensor:** Determines whether the soil requires watering.
2) **Temperature Sensor:** Measures environmental temperature, which can influence water requirements.
3) **Rain Sensor:** Detects rainfall and helps prevent unnecessary irrigation.
4) **Humidity Sensor:** Measures air humidity and can support irrigation scheduling.
5) **Water Level Sensor:** Monitors the water level in a storage tank to help prevent the pump from running dry.

Actuators Required:

1) **Water Pump:** Pumps water from a tank, well, or other water source.
2) **Solenoid Valve:** Opens or closes the water supply to a particular irrigation zone.
3) **Relay Module or Motor Driver:** Provides the required switching or control interface between the microcontroller and the pump or valve.
4) **Motorized Valve:** Controls water distribution across different field sections.

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5) **Water Level Sensor:** Monitors the water level in a storage tank to help prevent the pump from running dry.

Actuators Required:

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3) **Relay Module or Motor Driver:** Provides the required switching or control interface between the microcontroller and the pump or valve.
4) **Motorized Valve:** Controls water distribution across different field sections.

Example: When soil moisture falls below the set threshold and no rain-related condition prevents watering, the controller activates the pump. Once the required moisture level is reached, it stops irrigation.

*****ALL THE BEST*****